A model of geometric K-homology using groupoids

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1 Γ-CW proper complexes

Cita de lo que es un Γ-CW complex

Definition 1.1. *A Γ-CW complex is proper when...*

Definition 1.2. *A group is said to acts freely ans properly discontinuously on X if each element for all $g \neq e$ hold that given any $x \in X$ there exists an open set U in X such that $x \in U$ and $(gU) \cap U = \emptyset$.*

Remark 1.3. *The equivalence between conditions 1. and 2. in Definition* **??** *is obtained in Theorem 21 in [tD87].*

Revisar si se hace para grupos cualesquera

Definition 1.4. *The following are definitions of a model of a classifying space for proper actions of a group:*

1. *([BCH94], pg 6) A universal example for proper actions of G, denoted $\underline{E}G$, is a proper G-space with the following property: If X is any proper G-space, then there exists a G-map $\phi_X : X \to \underline{E}G$, and any two G-maps from X to $\underline{E}G$ are G-homotopic.*

2. *([DL17], p. 131) For a family of finite subgroups of G, the classifying space for proper actions $\underline{E}G$ is a proper $G - CW$ complex such that the fixed point set $\underline{E}G^H$ is contractible for every $H \in \mathcal{F}$.*

3. *A model of $\underline{E}\Gamma$ is a Γ-proper space with fixed points relative to subgroups of Γ are contractibles.*

Definition 1.5. *A Γ-vector bundle $p : E \to X$ over a Γ-space X is a vector bundle $p : E \to X$ such that, E is a Γ-space and p is a Γ-equivariant map.*

Remark 1.6. *For a Γ-vector bundle $p : E \to X$ over a proper Γ-space X, it satisfies that E is a Γ-proper space: consider a compact set $K \subset E$, then $p(K)$ is a compact set in X, and by the Γ-proper structure of X, we get that the set*

$$B := \{g \in \Gamma : g\ p(K) \cap p(K) \neq \emptyset\}$$

is finite. Let $A := \{g \in \Gamma : g\ K \cap K \neq \emptyset\}$. We have that $A \subseteq B$, because if $g \in A$, then $\emptyset \neq p(gK \cap K) \subseteq p(gK) \cap p(K) = gp(K) \cap p(K)$, where the last equality is because p is Γ-equivariant.

Definition 1.7. *For X, Y proper Γ-spaces we construct the fiber proper product $X \times_{\underline{E}\Gamma} Y$ as the Γ-equivariant fiber product (or equivariant pullback) of*

$$\phi_X : X \to \underline{E}\Gamma \leftarrow Y : \phi_Y.$$

Lemma 1.8. *([LO01], p. 599) If Γ is a discrete group and $f : X \to Y$ is an equivariant map between finite proper Γ-CW complexes, and $E \to X$ is a Γ-vector bundle. Then There is a Γ-vector bundle $F \to Y$ such that E is summand of f^*F.*

Remark 1.9. *We list two relevant observations about equivariant proper spaces.*

1. *If X is Γ-proper space, and Γ acts trivially on Y, then $X \times Y$ is a Γ-proper space.*

2. *By Remark 1.6. if X is Γ-proper space, $E \to X$ is a vector bundle, then $E \oplus \mathbb{R}^n$ is a proper Γ-vector bundle, where Γ acts trivialy over the real components.*

1.0.1 Groupoids

In this section we discuss about groupoids and groupoid actions where our main reference will be [EM10]. A **groupoid** is a small category such that any arrow is an isomorphism. In this way a groupoid $\mathcal{G}$ is compossed by a class of objects Z and a class of morphisms denoted $\mathcal{G}$ too. Now we present some spaces with groupoid actions.

Definition 1.10. *Let $\mathcal{G}$ be a topological groupoid*[1]*.*

- *A topological space X is called a* **$\mathcal{G}$-space** *if there is a continuous map $a : X \to Z$ called the* **anchor** *map and a homeomorphism*

$$\mathcal{G}_s \times_a X \to \mathcal{G}_t \times_a X : (g, x) \mapsto (g, g \cdot x),$$

  which determines the **action***, and holds associativity and unitality conditions.*

- *Definir grupoide propio teneindo en cuenta $\Gamma \ltimes X$*

Example 1.11. *For a group Γ, consider the groupoid $\mathcal{G} := \Gamma^*$, where $\mathcal{G}^0 := \{*\}$ and $\mathcal{G}^1 := \Gamma$.*

Example 1.12. *The transformation groupoid $\mathcal{G} \ltimes X$ (see [EM10], p. 4, Definition 2.2) is defined by X as its object space, morphisms the elements of $\mathcal{G}_s \times_a X$; the range and source maps are defined by $r(g, x) = g \cdot x$ and $s(g, x) = x$ respectively ; and its composition is given by $(g, x) \cdot (h, y) = (gh, y)$.*

[1] A topological groupoid is a groupoid $\mathcal{G}$ such that Z and $\mathcal{G}$ are Hausdorff and the structure maps source and target $s, t : \mathcal{G} \to Z$ are continuous and open (see [FSW20], p. 2).

Definition 1.13. *A $\mathcal{G}$-space X is called* **proper** *if the transformation groupoid $\mathcal{G} \ltimes X$ is a proper groupoid.*

Definition 1.14. *[EM10], p. 9 A properly numerable ver como podemos quitar la palabra numerable $\mathcal{G}$-space $E\mathcal{G}$ is* **universal** *if and only if, any properly numerable $\mathcal{G}$-space X admits a $\mathcal{G}$-map $X \to E\mathcal{G}$ and if there are two $\mathcal{G}$-maps $f, g : X \to E\mathcal{G}$, then f and g are $\mathcal{G}$-equivariantly homotopic.*

With this notion of universal space we have that:

1. If $\mathcal{G}$ is a locally compact groupoid, then there exist a construction of $E\mathcal{G}$.

Remark 1.15. *Let Γ be a group*

1. *If X is a Γ-space, it coincides with the notion of X being a Γ^*-space as a space with a groupoid action.*

2. *X is a Γ-space if and only if X is a $(\Gamma^* \ltimes X)$-space: If X is a Γ-space, define for $x \in X$ and for $(\gamma, y) \in \Gamma \times X$ such that $s(\gamma, y) = x$ (by the fact that the anchor map is the identity of X),*

$$(\gamma, x)x := r((\gamma, x)) = \gamma x;$$

Conversely, if X is a $(\Gamma^* \ltimes X)$-space we define

$$\gamma x := (\gamma, x)x.$$

3. *If X is a Γ-space, and considering the transform groupoid $\Gamma^* \ltimes X$, we get that for any $x \in X$, $\Gamma_x \cong (\Gamma \ltimes X)_x$, this because*

$$\begin{aligned}(\Gamma \ltimes X)_x &= \{(\gamma, x) \in \Gamma^*_s \times_a X : r((\gamma, x)) = x = s((\gamma, x))\} \\ &= \{(\gamma, x) \in \Gamma \times X : \gamma \cdot x = x\} = \Gamma_x\end{aligned}$$

4. *We say that a Γ-space X is proper if the transformation groupoid $\Gamma^* \ltimes X$ is a proper groupoid. When Γ is a discrete group, Parece definicion hay que demostrarlo.*

5. *$\underline{E}\Gamma(\Gamma \ltimes \underline{E}\Gamma) \cong_h \underline{E}\Gamma$ as $\Gamma \ltimes \underline{E}\Gamma$.*

2 Embedding theorems

In order to present and understand the *Factorization Theorem* (see [EM10], p. 18) then, we have to give some preliminaries. First, we remember the following definitions.

Definition 2.1. *Let $\mathcal{G}$ be a topological groupoid.*

- *A* **$\mathcal{G}$-vector bundle** *over a $\mathcal{G}$-space X is a vector bundle with $\mathcal{G}$-action such that the bundle projection, addition and scalar product are $\mathcal{G}$-equivariant.*

- *The* **pull back** *of a $\mathcal{G}$-vector bundle E over Z along the anchor map $a : X \to Z$ to obtain a $\mathcal{G}$-vector bundle over the $\mathcal{G}$-space X is denoted by E^X, and defined by*

$$E^X := X \times_Z E = X_a \times_{\pi_E} E = \{(x, e) \in X \times Z : a(x) = \pi_Z(e)\}$$

  A $\mathcal{G}$-vector bundle is called **trivial** *if it is isomorphic to E^X for some $\mathcal{G}$-vector bundle E over Z.*

  A $\mathcal{G}$-vector bundle F is called **subtrivial** *if it is a direct summand of a trivial $\mathcal{G}$-vector bundle (i.e., if there exist a $\mathcal{G}$-vector bundle $P \to X$ such that $F \oplus P \cong E^X$).*

Remark 2.2. *When $\mathcal{G}$ is the transformation groupoid (or action groupoid) of a group, we recover the notion of a Γ-vector bundle. For Γ be a discrete group, and X be a Γ-space, there is a bijection:*

$$\{\Gamma-\text{vector bundles over } X\} \leftrightarrow \{(\Gamma^* \ltimes X)-\text{vector bundles over } X\}$$

$(\leftarrow)$: *For a $(\Gamma^* \ltimes X)$-vector bundle $\rho : E \to X$, we note the Γ-vector bundle structure with action on E given by*

$$\gamma e := (\gamma, \rho(e))e,$$

where this action and Remark 1.15 make ρ a Γ-equivariant map

$$\rho(\gamma e) = \rho((\gamma, \rho(e))e) = (\gamma, \rho(e))\rho(e) = \gamma\rho(e).$$

$(\rightarrow)$: *If $\rho : E \to X$ is a Γ-vector bundle we note that it has a $(\Gamma^* \ltimes X)$-vector bundle structure by the anchor map $a := \rho : E \to X$, and the natural action*

$$(\gamma, x)e := \gamma e \quad \text{when } s((\gamma, x)) = a(e), \text{ i.e., } e \in E_x = \rho^{-1}(x)$$

where this implies that ρ is a $(\Gamma^* \ltimes X)$-equivariant map by

$$\rho((\gamma, x)e) = \rho(\gamma e) = \gamma\rho(e) = \gamma x \quad \Leftrightarrow \quad (\gamma, x)e \in E_{\gamma x} = \rho^{-1}(\gamma x).$$

Definition 2.3 ([EM10], p. 10)**.** *Let $\mathcal{G}$-be a topological groupoid and let X be a numerably proper $\mathcal{G}$-space.*

- *There are* **enough $\mathcal{G}$-vector bundles** *on X if for every $x \in X$ and every finite dimensional representation*[2] *of the stabilizer $\mathcal{G}_x^x$, there is a $\mathcal{G}$-vector bundle over X whose fibre at x contains the given representation of $\mathcal{G}_x^x$.*

- *A $\mathcal{G}$-vector bundle V on X is* **full** *if for every $x \in X$, the fibre V_x contains all irreducible representations of the stabilizer $\mathcal{G}_x^x$.*

Remark 2.4. *We list some important facts that will be useful about the proof of Theorem 2.5 in [EM10].*

[2] An introduction on representations of groups can be consulted in [Ser77].

1. *The vector bundle E^W is the $\mathcal{G}$-vector bundle with total space $Y \times_Z E$, which coincides with the pullback through the maps a and the bundle projection ρ_E:*

$$\begin{tikzcd} E^W := Y \times_Z E \arrow[r] \arrow[d] & E \arrow[d, "\rho_E"] \\ Y \arrow[r, "a"'] & Z \end{tikzcd}$$

   This means that if $Z = \{*\}$, then any vector bundle $E \to Z$ must be trivial, $E = \{*\} \times \mathbb{R}^n$, and therefore $E^Y = Y \times_Z E = Y \times (\{*\} \times \mathbb{R}^n) \cong Y \times \mathbb{R}^n$ is also a trivial bundle over Y.

2. *The $\mathcal{G}$-vector bundle $E \to Z$ is picked such that we can get a fibrewise smooth embedding $X \hookrightarrow E$ (this is guaranteed by Theorem 3.22 in [EM10] when X is a smooth $\mathcal{G}$-manifold, such that $\mathcal{G}\backslash X$ has finite covering dimension and there is a full $\mathcal{G}$-vector bundle on Z).*

$$X \hookrightarrow E \qquad \Rightarrow \qquad \begin{tikzcd} & E^Y = a^*E \arrow[d] & E \arrow[d] \\ X \arrow[r, "f"] & Y \arrow[r, "a"] & Z \end{tikzcd}$$

   Note that the choice of E is independent of the space Y.

3. *The total space V of the vector bundle $V \to X$ is the normal bundle*[3] *$\nu_{f \times g}$ of the embedding $(f \times g) : X \to E^Y$ constructed using the embedding $g : X \hookrightarrow E$.*

Now, we present an argument that let us to use the theory of Emmerson - Meyer, mainly Theorem 2.5 in the principal objects of our problem of interest. First we present an important result of Luck.

reparafrasearlo con grupoide $\Gamma \ltimes \underline{E}\Gamma$

Theorem 2.5 ([EM10], p. 18)**.** *Let $\mathcal{G}$ be a proper groupoid, let X and Y be smooth $\mathcal{G}$-manifolds, and let $f : X \to Y$ be a smooth $\mathcal{G}$-equivariant map. Suppose that*

- *Z has enough $\mathcal{G}$-vector bundles and $\mathcal{G}\backslash X$ is compact.*

Then, there are

- *a smooth $\mathcal{G}$-vector bundle V over X,*
- *a smooth $\mathcal{G}$-vector bundle E over Z,*
- *a smoth $\mathcal{G}$-equivariant, open embedding $\eta_f : V \to E^Y$,*

[3] For an introduction to normal bundles, see [GP10].

such that $f = \rho_{E^Y} \circ \eta_f \circ \xi_V$, where $\xi_V : X \to V$ is the zero-section of the fiber bundle $\rho_V : V \to X$.

$$\begin{tikzcd} V \arrow[r, "\eta_f"] \arrow[d, "\rho_V"'] & E^Y \arrow[d, "\rho_{E^Y}"] & E \arrow[d] \\ X \arrow[r, "f"'] & Y \arrow[r, "a"'] & Z \end{tikzcd}$$

3 Useful constructions

3.1 Connected sum generalization

In [Kos92], p. 99, we found a generalization of connected sum of manifolds called the *joining manifolds along submanifolds with boundary* (also called the *pasting*) which is as follows: Consider

1. M_1 and M_2 are two $(n+k)$-dimensional smooth manifolds,
2. $E \to N$ being a k-dimensional Riemannian vector bundle over the n-dimensional closed compact submanifold N,
3. define $\alpha_E : E \setminus \xi_E(N) \to E \setminus \xi_E(N)$ by $\alpha_E(||v||) = \alpha(v) \cdot \frac{v}{||v||}$, where $\alpha : (0,\infty) \to (0,\infty)$ an orientation reversing diffeomorphism,
4. $h_i : E \to M_i$ are Γ-embeddings such that $h_i|_{\xi_E(N)}$ is a neat embedding where $h_i(E)$ are neat tubular neighborhoods of $h_i(\xi_E(N))$.

With these, we define $M(h_1, h_2)$ the join manifold along N as $M_1 \cup M_2/\sim$, where for $v \in h_1(E \setminus \xi_E(N))$ we consider the identification $v \sim h_2 \circ \alpha_E \circ h_1^{-1}(v)$. We get that $M(h_1, h_2)$ is an oriented smooth manifold with boundary (see [Kos92], p. 100).

Remark 3.1. *Recall that if $\rho : E \to W$ is a Γ-vector bundle, then $TE \cong TW \oplus E$: we have the exact sequence $0 \to \rho^*E \to TE \to \rho^*TW \to 0$ (see [Spi99], p. 103), then the restricted exact sequence $0 \to \rho^*E|_W \to TE|_W \to \rho^*TW|_W \to 0$, where $W \subseteq E$ by the zero section s, is identified with $0 \to E \to TE|_W \to TW \to 0$, by the fact that $\rho^*E|_W \cong E$ and $(\rho^*TW)|_W \cong TW$. We have that this sequence split, because the map $d\rho : TE|_W \to TW$ has right inverse by $\rho \circ s = id_W$.*

3.2 Deformation to the normal cone

Let be M a manifold and $N \subseteq M$ a submanifold of M. The deformation to the normal cone of N in M (see [DS19]) is

$$DNC(M, N) = (M \times \mathbb{R}^*) \sqcup (\nu_N^M \times \{0\})$$

as set and with smooth structure given by the diffeomorphism (onto its image) $\Theta : W' \to M \times \mathbb{R}$, where $W' = \{(x, \xi, t) \in \nu_N^M \times \mathbb{R} : (x, t\xi) \in U'\}$ is an

open neighbourhood of $\nu_N^M \times \{0\}$ in $\nu_N^M \times \mathbb{R}$, and where Θ is defined using an exponential map $\theta : U' \subseteq \nu_N^M \to U \subseteq M$ as

$$\Theta(x,\xi,t) = \begin{cases} (\theta(x,t\xi),t) & \text{for } t \neq 0 \\ (x,\xi,0) & \text{for } t = 0 \end{cases}$$

Example 3.2. *Consider a Γ-equivariant smooth map $f : \partial M \to W$ for M and W Γ-manifolds with boundary. By Theorem 2.5, we have the commutative diagram*

$$\begin{tikzcd} V \arrow[r, "\eta_f"] & E \arrow[d, "\rho_E"] \\ \partial M \arrow[u, "\xi_{\partial M}"] \arrow[r, "f"'] \arrow[ur, "\tilde{g}"] & W \end{tikzcd}$$

where η_f is a Γ-embedding. Then in this case

$$DNC(E,\partial M)|_{[0,1]} = (E \times (0,1]) \sqcup (\nu_{\partial M}^E \times \{0\}) = (E \times (0,1]) \sqcup (V \times \{0\}),$$

where $\partial(DNC(E,\partial M)|_{[0,1]}) = (V \times \{0\}) \cup (E \times \{1\}) \cup (E|_{\partial W} \times (0,1))$ Podemos ponerlo un poco mas general.

4 Pushforward homology

Definition 4.1 ([Lüc02])**.** *A Γ-homology theory $\mathscr{H}_*^\Gamma$ with values in R-modules is a collection of covariant functors $\mathscr{H}_n^\Gamma$ from the category of Γ-proper CW cocompact pairs to the category of R-modules, indexed by $n \in \mathbb{Z}$, together with natural transformations called the boundary maps*

$$\partial_n^\Gamma : \mathscr{H}_n^\Gamma(X,A) \to \mathscr{H}_{n-1}^\Gamma(A)$$

for $n \in \mathbb{Z}$ (where $\mathscr{H}_*^\Gamma(A) := \mathscr{H}_*^\Gamma(A,\emptyset)$), such that the following axioms are satisfied:

1. ***Γ-homotopy invariance.***
   If f and g are proper Γ-homotopic maps $(X,A) \to (Y,B)$ of proper Γ-pairs, then the induced maps

   $$f_*, g_* : \mathscr{H}_n^\Gamma(X,A) \to \mathscr{H}_n^\Gamma(Y,B)$$

   are the same for all $n \in \mathbb{Z}$.

2. ***Long exact sequence of a pair.***
   Given a proper Γ-CW cocompact pair (X,A), there is a long exact sequence

   $$\ldots \xrightarrow{j_*} \mathscr{H}_{n+1}^\Gamma(X,A) \xrightarrow{\partial_{n+1}^\Gamma} \mathscr{H}_n^\Gamma(A) \xrightarrow{i_*} \mathscr{H}_n^\Gamma(X) \xrightarrow{j_*} \mathscr{H}_n^\Gamma(X,A) \xrightarrow{\partial_n^\Gamma} \ldots,$$

   where $i : A \to X$ and $j : X \to (X,A)$ are the inclusions.

3. ***Excision.***

   Let (X, A) be a Γ-CW cocompact proper pair and let $f : A \to B$ be a cellular Γ-map of proper Γ-CW cocompact pairs. Equip $(X \cup_f B, B)$ with the induced structure of a proper Γ-CW cocompact pair. Then the canonical map $F : (X, A) \to (X \cup_f B, B)$ induces an isomorphism

$$F_* : \mathscr{H}_n^\Gamma(X, A) \to \mathscr{H}_n^\Gamma(X \cup_f B, B).$$

4. ***Disjoint union axiom.***

   Let $\{X_i \mid i \in I\}$ be a family of proper Γ-CW-complexes. Denote by

$$j_i : X_i \to \coprod_{i \in I} X_i$$

   the canonical inclusion. Then the map

$$\bigoplus_{i \in I} j_i^* : \bigoplus_{i \in I} \mathscr{H}_\Gamma^n(X_i) \to \mathscr{H}_\Gamma^n(\coprod_{i \in I} X_i)$$

   is a group isomorphism.

Definition 4.2 ([Lüc05])**.** *A Γ-cohomology theory $\mathscr{H}_\Gamma^*$ with values in R-modules is a collection of covariant functors $\mathscr{H}_\Gamma^n$ from the category of Γ-pairs to the category of R-modules, indexed by $n \in \mathbb{Z}$, together with natural transformations called the boundary maps*

$$\delta_\Gamma^n : \mathscr{H}_\Gamma^n(A) \to \mathscr{H}_\Gamma^{n+1}(X, A)$$

for $n \in \mathbb{Z}$, such that the following axioms are satisfied:

1. ***Γ-homotopy invariance.***

   If f_0 and f_1 are proper Γ-homotopic maps $(X, A) \to (Y, B)$ of proper Γ-pairs, then the induced maps

$$f_0^n, f_1^n : \mathscr{H}_\Gamma^n(X, A) \to \mathscr{H}_\Gamma^n(Y, B)$$

   are the same for all $n \in \mathbb{Z}$.

2. ***Long exact sequence of a pair.***

   Given a proper Γ-pair (X, A), there is a long exact sequence

$$\ldots \xrightarrow{\delta^{n-1}} \mathscr{H}_\Gamma^n(X, A) \xrightarrow{j^n} \mathscr{H}_\Gamma^n(X) \xrightarrow{i^n} \mathscr{H}_\Gamma^n(A) \xrightarrow{\delta_\Gamma^n} \mathscr{H}_\Gamma^{n+1}(X, A) \xrightarrow{j^{n+1}} \ldots,$$

   where $i : A \to X$ and $j : X \to (X, A)$ are the inclusions.

3. ***Excision.***

   Let (X, A) be a Γ-CW cocompact properpair and let $f : A \to B$ be a cellular Γ-map of proper Γ-pairs. Equip $(X \cup_f B, B)$ with the induced structure of a proper Γ-pair. Then the canonical map $F : (X, A) \to (X \cup_f B, B)$ induces an isomorphism

$$F^n : \mathscr{H}_\Gamma^n(X \cup_f B, B) \to \mathscr{H}_\Gamma^n(X, A).$$

4. ***Disjoint union axiom.***
Let $\{X_i \mid i \in I\}$ be a family of proper Γ-CW-complexes. Denote by

$$j_i : X_i \to \coprod_{i \in I} X_i$$

the canonical inclusion. Then the map

$$\prod_{i \in I} j_i^* : \mathscr{H}_\Gamma^n(\coprod_{i \in I} X_i) \to \prod_{i \in I} \mathscr{H}_\Gamma^n(X_i)$$

is a group isomorphism.

Definition 4.3. *A Γ-cohomology theory $\mathscr{H}_\Gamma^*$ is called multiplicative if holds:*

1. *For any proper Γ-pair (X, A), it satisfy for C_A, a direct set of closed Γ-invariant neighborhoods of A, that*

$$\mathscr{H}_\Gamma^*(X, A) \cong \varinjlim_{U \in C_A} \mathscr{H}_\Gamma^*(X, U).$$

2. *$\mathscr{H}_\Gamma^*$ is endowed with a graded commutative exterior product*

$$\begin{array}{c} \mathscr{H}_\Gamma^i(X_1, A_1) \otimes \mathscr{H}_\Gamma^j(X_2, A_2) \\ \downarrow \\ \mathscr{H}_\Gamma^{i+j}(X_1 \times_{\underline{E}\Gamma} X_2, X_1 \times_{\underline{E}\Gamma} A_2 \cup A_1 \times_{\underline{E}\Gamma} X_2) \end{array}$$

for all $i, j \in \mathbb{Z}$, which are compatible with the boundary maps.

3. *There is a $1 \in \mathscr{H}_\Gamma^0(\underline{E}\Gamma)$ such that for every pair (X, A) and any $\eta \in \mathscr{H}_\Gamma^i(X, A)$, $1 \otimes \eta = \eta \otimes 1 = \eta$.*

Definition 4.4. *For a d-dimensional real Γ-vector bundle $E \to X$ over a proper Γ-space X, a $\mathscr{H}_\Gamma$-orientation for E is a class $\tau \in \mathscr{H}_\Gamma^d(E)$ such that for each proper Γ-map $f : Y \to X$, multiplication with $f^*\tau \in \mathscr{H}_\Gamma^d(f^*E)$ induces an isomorphism (refered as Thom-isomorphism)*

$$- \otimes \tau : \mathscr{H}_\Gamma^*(Y) \to \mathscr{H}_\Gamma^{*+d}(f^*E).$$

We say that a smooth manifold with boundary M is a $\mathscr{H}_\Gamma$-orientable if TM is $\mathscr{H}_\Gamma$-orientable. When is chosen a fixed $\mathscr{H}_\Gamma$-orientation for a real Γ-vector bundle (or particularly for a manifold), we say that it is a $\mathscr{H}_\Gamma$-oriented bundle (or manifold) Como arregla Luck la orientacion para pareja. Definir cohmologia de fibrado.

Definition 4.5. *Nueva definicion para lograr compacidad For a d-dimensional real Γ-vector bundle $E \to X$ over a proper Γ-space X, a $\mathscr{H}_\Gamma$-orientation for E is a class $\tau \in \mathscr{H}_\Gamma^d(S(E \oplus \mathbb{R}))$ such that for each proper Γ-map $f : Y \to X$,*

$$\begin{tikzcd}
f^*S(E\oplus\mathbb{R}) \arrow[d] & S(E\oplus\mathbb{R}) \arrow[r, hook] \arrow[d] & E\oplus\mathbb{R} \arrow[dl] \\
Y \arrow[r, "f"] & X &
\end{tikzcd}$$

multiplication with $f^*\tau \in \mathscr{H}^d_\Gamma(f^*S(E\oplus\mathbb{R}))$ induces an isomorphism (refered as Thom-isomorphism)

$$-\otimes\tau : \mathscr{H}^*_\Gamma(Y) \to \mathscr{H}^{*+d}_\Gamma(f^*S(E\oplus\mathbb{R})).$$

We say that a smooth manifold with boundary M is a $\mathscr{H}_\Gamma$-orientable if TM is $\mathscr{H}_\Gamma$-orientable. When is chosen a fixed $\mathscr{H}_\Gamma$-orientation for a real Γ-vector bundle (or particularly for a manifold), we say that it is a $\mathscr{H}_\Gamma$-oriented bundle (or manifold).

Decir que usamos teoria de cohomologia de soporte compacto. Ojala incluir cita para indicar que significa esto

Remark 4.6. *Note that in conditions of Definition 4.4, if we run the multiplicative structure in order to get the map by multiplication of τ, we need that*

$$\mathscr{H}^{*+d}_\Gamma(Y\times_{\underline{E}\Gamma} E) \cong \mathscr{H}^{*+d}_\Gamma(f^*E)$$

Idea:

$$\begin{tikzcd}
Y\times_{\underline{E}\Gamma} E \arrow[d, "\rho_Y"'] & E \arrow[d, "\rho_E"'] \\
Y \arrow[r, "f"] \arrow[dr, "a_Y"'] & X \arrow[d, "a_X"] \\
& \underline{E}\Gamma
\end{tikzcd}$$

We can consider the model of $\underline{E}\Gamma \cong CX\times\underline{E}\Gamma$ (where the structure of Γ-CW of this space is....) tenemos problemas con punto fijo en CX, utilizar en reemplazo $(X\times[0,1], X\times\{1\})$. With this model we recall that

$$\begin{aligned} Y\times_{\underline{E}\Gamma} E &= \{(y,e)\in Y\times E : a_Y(y) = a_X\circ f(y) = a_X\circ\rho_E(e)\}, \\ f^*E &= \{(y,e)\in Y\times E : f(y) = \rho_E(e)\}. \end{aligned}$$

We denote vector dimension of real smooth vector bundles by $dim_{\mathbb{R}}$, and denote manifold dimension by dim, without the subscript.

Definition 4.7. *A Pushforward structure over a multiplicative Γ-cohomology theory $\mathscr{H}^*_\Gamma$ is an assign for any proper Γ-equivariant, orientation preserving smooth map $f : M\to N$ of manifolds with boundary, a homomorphism*

$$f! : \mathscr{H}^*_\Gamma(M) \to \mathscr{H}^{*+dim(N)-dim(M)}_\Gamma(N),$$

such that

1. *For a composable proper Γ-equivariant, orientation preserving smooth maps we have that $(f \circ g)! = f! \circ g!$ and $Id_X! = Id_{\mathscr{H}^*_\Gamma(X)}$.*

2. *If $F : \mathbb{R} \times M \to N$ is a smooth Γ-equivariant orientation preserving map, then $F(0,-)! = F(1,-)!$.*

3. *If $E \to M$ is a smooth real $\mathscr{H}_\Gamma$-oriented Γ-vector bundle over a manifold with boundary M, then if $s : M \to E$ is the zero section, the pushforward map*

$$(s \oplus 1)! : \mathscr{H}^*_\Gamma(M) \to \mathscr{H}^{*+dim(E)-dim(M)}_\Gamma(S(E \oplus \mathbb{R}))$$

is the Thom isomorphism described in Definition 4.4 Note que no cambiamos la dimension del codominio, pues dim(S(E+R))=dim(E).

4. *Let N be a manifold with boundary and consider $N^0 = N \setminus \partial N$. If $i^0 : N^0 \to N$ the inclusion, and if U is an open collar around ∂N in N and let $l : \partial N \to U \to N^0$, the inclusion through the collar, then the pushforwards $i^0!$ and $l!$ corresponds to the maps appearing in the long exact sequence of the pair $(N, \partial N)$,*

$$\cdots \to \mathscr{H}^{*-1}_\Gamma(\partial N) \xrightarrow{l!} \mathscr{H}^*_\Gamma(N^0) \xrightarrow{i^0!} \mathscr{H}^*_\Gamma(N) \to \cdots$$

Here we are using the identification $\mathscr{H}^*_\Gamma(N^0) \cong \mathscr{H}^*_\Gamma(N, \partial N)$. Then, we mean that for the long exact sequence for the maps $i : \partial N \to N$ and $j : N \to (N, \partial N)$, we get that $l! = \delta^{*-1}$ and $i^0! = j^*$.

5. *For any $f : (M, \partial M) \to (N, \partial N)$, we had that for the commutative diagram*

$$\begin{tikzcd} M \arrow[r, "f"] & N \\ \partial M \arrow[u, "i_{\partial M}"] \arrow[r, "f|_{\partial M}"'] & \partial N \arrow[u, "i_{\partial N}"'] \end{tikzcd}$$

it is satisfied that $i^*_{\partial N} \circ f! = (f|_{\partial M})! \circ i^*_{\partial M}$.

6. *If $f : M_1 \to N_1$ and $g : M_2 \to N_2$, then for the map*

$$(f \sqcup g) : M_1 \sqcup M_2 \to N_1 \sqcup N_2,$$

we have that

$$(f \sqcup g)! = f! \oplus g! : \mathscr{H}^*_\Gamma(M_1) \oplus \mathscr{H}^*_\Gamma(M_2) \to \mathscr{H}^*_\Gamma(N_1) \oplus \mathscr{H}^*_\Gamma(N_2)$$

7. *If $f_i : M_i \to N$ and $+$ denotes the operation in $\mathscr{H}^*(N)$, then for the map $f_1 \cup f_2 : M_1 \sqcup M_2 \to N$ defined such that $(f_1 \cup f_2) \circ i_k = f_k$, then $(f_1 \cup f_2)! = f_1! + f_2!$.*

8. *If $i_M : M \to M \sqcup N$ is the natural inclusion, then*

$$i_M! : \mathscr{H}^*_\Gamma(M) \to \mathscr{H}^*_\Gamma(M \sqcup N) \cong \mathscr{H}^*_\Gamma(M) \oplus \mathscr{H}^*_\Gamma(N) : x \mapsto x \oplus 0.$$

Definition 4.8. *Let $\mathscr{H}^*_\Gamma$ be a multiplicative Γ-cohomology theory with a pushforward structure. Let (X, A) be a proper Γ-CW pair, let $n \in \mathbb{Z}$, we define the abelian group $\mathscr{H}^{pf}_*(X, A; \Gamma)$ as the set of equivalence relations of tuples $(M, \partial M, \pi, x)$ (called* **cycles***), where:*

1. *$(M, \partial M)$ is a Γ-proper manifold possibly with boundary with a $\mathscr{H}_\Gamma$-orientation.*

2. *$\pi_M : (M, \partial M) \to (X, A)$ is a continuous Γ-equivariant map such that $\pi_M(\partial M) \subseteq A$.*

3. *$x_M \in \mathscr{H}^n_\Gamma(M)$ such that $n = * + dim(M)$. If $(M, \partial M)$ is no a cocompact pair, then $(M, \partial M, \pi_M, x_M)$ is a cycle if and only if $x_M \in \mathscr{H}^n_\Gamma(M) := \mathscr{H}^n_\Gamma(\tilde{M}, \tilde{M} \setminus M)$, where $(\tilde{M}, \partial\tilde{M})$ is a cocompact Γ-pair that contains $(M, \partial M)$ embedded.*

4. *Two cycles $(M, \partial M, \pi_M, x_M)$ and $(N, \partial N, \pi_N, x_N)$ are isomorphic cycles if there exist a diffeomorphism $\phi : M \to N$ such that $\pi_M = \pi_N \circ \phi$ and $\phi^*(x_N) \cong x_M$.*

5. *The equivalence relation*

$$(M, \partial M, \pi_M, x_M) \sim (N, \partial N, \pi_N, x_N),$$

is given if there is a k-cycle $(W, \partial W, \pi_W, x_W)$ and proper smooth, Γ-equivariant, $\mathscr{H}_\Gamma$-orientation preserving maps $f_M : (M, \partial M) \to (W, \partial W)$ and $f_N : (N, \partial N) \to (W, \partial W)$ such that, $\pi_M = \pi_W \circ f_M$, $\pi_N = \pi_W \circ f_N$ and $f_M!(x_M) = x_W = f_N!(x_N) \in \mathscr{H}^*_\Gamma(W)$, as shown in the following diagram

$$\begin{tikzcd}
 & (W, \partial W) \arrow[dd, "\pi_W"] & \\
(M, \partial M) \arrow[ur, "f_M"] \arrow[dr, "\pi_M"'] & & (N, \partial N) \arrow[ul, "f_N"'] \arrow[dl, "\pi_N"] \\
 & (X, A) &
\end{tikzcd}$$

6. *The operation of cycles $[M, \partial M, \pi, x] + [N, \partial N, \pi', x'] \in \mathscr{H}^{pf}_*(X, A; \Gamma)$ is defined by*

$$[M \sqcup N, \partial M \sqcup \partial N, \pi \sqcup \pi', x \oplus x'],$$

where $M \sqcup N$ has the induced $\mathscr{H}_\Gamma$-orientation and the cohomology element $x \oplus y \in \mathscr{H}^{*+dim(M)}_\Gamma(M) \bigoplus \mathscr{H}^{*+dim(N)}_\Gamma(N)$.

Remark 4.9. *If $[M, \partial M, \pi_1, x_M] \in \mathscr{H}_*^{pf}(X, A; \Gamma)$ and π_1 is Γ-homotopic to a map $\pi_2 : M \to X$. This is by the fact that there is a Γ-homotopy $F : M \times \mathbb{R} \to X$ such that $F(m, 0) = \pi_1$ and $F(m, 1) = \pi_2$ (with the triviql Γ-action on the second factor). Therefore we get the commutative diagram*

$$\begin{tikzcd}
 & M \times \mathbb{R} \arrow[dd, "F"] & \\
M \arrow[ur, "Id_M \times 0"] \arrow[dr, "\pi_1"'] & & M \arrow[ul, "Id_M \times 1"'] \arrow[dl, "\pi_2"] \\
 & X &
\end{tikzcd}$$

Since $Id_{M \times \mathbb{R}}$ is an homotopy between $Id_M \times 0$ and $Id_M \times 1$, then by the pushforward structure $(Id_M \times 0)!(X_M) = (Id_M \times 1)!(X_M)$.

Lemma 4.10. *If $(M, \partial M, \pi_M, x_M) \sim (N, \partial N, \pi_N, x_N)$ by the maps $f_M : M \to W$ and $f_N : N \to W$, then we get that $(M, \partial M, \pi_M, x_M) \sim (N, \partial N, \pi_N, x_N)$ by maps $\phi_M : M \to E^W$ and $\phi_N : N \to E^W$ with ϕ_N being an embedding and $E^W \to W$ is a vector bundle.*

Proof. Applying Theorem 2.5 to the map $f_N : N \to W$, we get that $f_N = \rho_{E^W} \circ \eta_{f_N} \circ \xi_V$ with $\eta_{f_N} : V \to E^W$ is an embedding,

$$\begin{tikzcd}
 & E^W \arrow[d, "\rho_{E^W}"'] & \\
 & W \arrow[dd, "\pi_W"] & V \arrow[ul, "\eta_{f_N}"'] \arrow[d, "\rho_V"] \\
M \arrow[ur, "f_M"] \arrow[dr, "\pi_M"'] & & N \arrow[ul, "f_N"'] \arrow[dl, "\pi_N"] \\
 & X &
\end{tikzcd}$$

where $\xi_V : N \to V$ denotes the zero section of the vector bundle $\rho : V \to N$. We want to show that $(M, \partial M, \pi_M, x_M) \sim (N, \partial M, \pi_N, x_N)$ through the vector bundle E^W rather than W, using a diagram

$$\begin{tikzcd}
 & E^W \arrow[dd, "\pi_{E^W}"] & \\
M \arrow[ur, "\phi_M"] \arrow[dr, "\pi_M"'] & & N \arrow[ul, "\phi_N"'] \arrow[dl, "\pi_N"] \\
 & X &
\end{tikzcd}$$

where, $\phi_M = \xi_{E^W} \circ f_M$, $\phi_N = \eta_{f_N} \circ \xi_V$ and $\pi_{E^W} = \pi_W \circ \rho_{E^W}$. We conclude the proof verifying the following six conditions in order to stablish the relation exhibited in the previous diagram.

- π_{E^W} is a continuous Γ-equivariant map, by composition of continuous Γ-equivariant maps.

- ϕ_M and ϕ_N are proper smooth, Γ-equivariant and $\mathscr{H}_\Gamma$-orientation preserving maps, because zero sections are (locally) smooth such as f_M and η_{f_N} by definition, composition of equivariant maps is equivariant too and $\mathscr{H}_\Gamma$-orientation preserving.

- ϕ_N is an embedding by composition of embeddings, because ξ_V is an embedding by the fact that $\rho_V \circ \xi_V = id_N$ implies that ξ_V is inyective and for any $n \in N$, the derivative $d_n\xi_V$ is inyective by the composition $d_{\xi_V(n)}\rho_V \circ d_n\xi_V = id_{T_nN}$.

- Since $\xi_{E^W} : W \to E^W$ is the zero section, we have that $\rho_{E^W} \circ \xi_{E^W} = id_W$, then $\rho_{E^W}! \circ \xi_{E^W}! = id_W!$ by the pushforward structure, where $\xi_{E^W}!$ is the Thom isomorphism mentioned in Definition 4.7, we get that $\xi_{E^W}! \circ \rho_{E^W}! = id_{E^W}!$, and since $f_N!(x_N) = \rho_{E^W}! \circ \eta_{f_N}! \circ \xi_V!(x_N)$ and $f_M!(x_M) = f_N!(x_N)$, we obtain

$$\begin{aligned}
\phi_N!(x_N) &= \eta_{f_N}! \circ \xi_V!(x_N) \\
&= \xi_{E^W}! \circ \rho_{E^W}! \circ \eta_{f_N}! \circ \xi_V!(x_N) \\
&= \xi_{E^W}! \circ f_N!(x_N) \\
&= \xi_{E^W}! \circ f_M!(x_M) \\
&= \phi_M!(x_M).
\end{aligned}$$

- We have that $\pi_{E^W} \circ \phi_M = \pi_W \circ (\rho_{E^W} \circ \xi_{E^W}) \circ f_M = \pi_W \circ f_M = \pi_M$.

- We have that $\pi_{E^W} \circ \phi_N = \pi_W \circ (\rho_{E^W} \circ \eta_{f_N} \circ \xi_V) = \pi_W \circ f_N = \pi_N$.

□

Remark 4.11. *Note that we can consider the embedding obtained in Lemma 4.10 as an embedding $N \to W'$, with $W' := S(TE^W \oplus \mathbb{R})$ a compact manifold, as the following commutative diagram induce*

$$\begin{tikzcd}
N \arrow[r, "\phi_N"] \arrow[dr, "\pi_N"'] & E^W \arrow[r, "s\oplus 1"] \arrow[d, "\pi_{E^W}"] & W' \arrow[dl, "\pi_{E^W} \circ \rho_{TE^W} \circ \pi_1"] \\
& X &
\end{tikzcd}$$

where $(s \oplus 1) \circ \phi_N$ is a composition of Γ-embeddings, and

$$\rho_{TE^W} \circ \pi_1 \circ (s \oplus 1) = id_{E^W}.$$

Lemma 4.12. *For any Γ-proper embeddings $i_1 : N \to W_1$ and $i_2 : N \to W_2$, there exist Γ-vector bundles $D_1 \to W_1$ and $D_2 \to W_2$ such that*

$$\nu_{\xi_{D_1} \circ i_1} \cong \nu_{\xi_{D_2} \circ i_2}.$$

Proof. Consider the tangent vector bundles $TW_k \to W_k$ for $k = 1, 2$ and the unique up to equivariant homotopy maps $a_1 : W_1 \to Z$, and $a_2 : W_2 \to Z$, by the universal property of $Z = \underline{E}\Gamma$. By the Lemma 1.8, we get that $a_1^* E_1 \cong TW_1 \oplus C_1$ and $a_2^* E_2 \cong TW_2 \oplus C_2$, for Γ-vector bundles $E_1, E_2 \to Z$, $C_1 \to W_1$ and $C_2 \to W_2$, as shows the following diagram:

$$\begin{tikzcd}
E_1 \arrow[d] & a_1^* E_1 \cong TW_1 \oplus C_1 \arrow[d] & & a_2^* E_2 \cong TW_2 \oplus C_2 \arrow[d] & E_2 \arrow[d] \\
Z & W_1 \arrow[l, "a_1"'] & N \arrow[l, "i_1"'] \arrow[r, "i_2"'] & W_2 \arrow[r, "a_2"'] & Z
\end{tikzcd}$$

(We have that Z is the universal proper Γ-CW complex (c.f. [MV03], p. 6) and, by Remark 4.11 we can consider W_i as a compact proper Γ-manifold, making possible the application of Lemma 1.8) If we consider the Γ-vector bundles $D_1 := a_1^* E_2 \oplus C_1$ and $D_2 := a_2^* E_1 \oplus C_2$ over W_1 and W_2, respectively. With this in mind, denoting $E := E_1 \oplus E_2$, we note that:

1. $TD_1|_N \cong TD_2|_N$: by one hand we have $TD_1 \cong TW_1 \oplus D_1$ by Remark 3.1, and by the other hand

$$\begin{aligned}
(a_1 \circ i_1)^* E &= i_1^*(a_1^* E_1 \oplus a_1^* E_2) \\
&= i_1^*(TW_1 \oplus C_1 \oplus a_1^* E_2) \\
&\cong i_1^*(TW_1 \oplus D_1) \\
&\cong TW_1|_N \oplus D_1|_N.
\end{aligned}$$

   Therefore, $TD_1|_N \cong (a_1 \circ i_1)^* E$, and analogously, $TD_2|_N \cong (a_2 \circ i_2)^* E$. By the universal property of Z, we get that the maps $a_1 \circ i_1, a_2 \circ i_2 : N \to Z$ are Γ-homotopical, then $TD_1|_N \cong TD_2|_N$.

2. $TN \leq TD_i|_N$: Since D_1 is a vector bundle over W_1, using the composition of its zero section and the map i_1, we find that there is an embedding $N \hookrightarrow D_1$, and therefore $TD_1|_N \cong TN \oplus \nu_N^{D_1}$, which implies $TN \leq TD_1|_N$.

Then, we get that $\nu_1 := \nu_N^{D_1} \cong TD_i|_N / TN \cong \nu_N^{D_2} =: \nu_2$. □

Remark 4.13. *For maps $i_1 : N \to W_1$ and $i_2 : N \to W_2$ being Γ-embeddings, by Lemma 4.12, there exist Γ-vector bundles $D_k \to W_k$ for $k = 1, 2$, such that $\nu_1 := \nu_N^{D_1} \cong TD_i|_N / TN \cong \nu_N^{D_2} =: \nu_2$. With this isomorphism $\phi : \nu_1 \to \nu_2$, we construct the manifold*

$$W := (D_1 \oplus \mathbb{R})_{\nu_1} \#_{\nu_2} (D_2 \oplus \mathbb{R}),$$

the joint of the vector bundles (manifolds) $D_1 \oplus \mathbb{R}$ and $D_2 \oplus \mathbb{R}$ along the submanifold N across the normal bundles $\nu_1 \oplus \mathbb{R}$ and $\nu_2 \oplus \mathbb{R}$, where

$$\partial W = [D_1|_{\partial W_1} \cup D_2|_{\partial W_2}] \oplus \mathbb{R} \setminus [\xi_{\nu_1} \circ i_1(N) \times \{1\}],$$

where ξ_{ν_k} is the zero section of the vector bundle ν_k. This is made by cutting the identified copies of N, $(\xi_{\nu_1} \circ i_1(N)) \times \{1\}$ and $(\xi_{\nu_2} \circ i_2(N)) \times \{1\}$, where this

identification is obtained by the map $(\xi_{\nu_2} \times \{1\} \circ i_2) \circ (\phi \times \{1\}) \circ (\xi_{\nu_1} \times \{1\} \circ i_1)^{-1}$*, making the following diagram commutative in* W*, for any* $r \in \mathbb{R} \setminus \{1\}$*:*

$$\begin{tikzcd}
D_1 \oplus \mathbb{R} & & D_2 \oplus \mathbb{R} \\
\nu_1 \times \mathbb{R} \arrow[u, hook] \arrow[rr, "\phi \times Id_{\mathbb{R}}"] & & \nu_2 \times \mathbb{R} \arrow[u, hook] \\
& N \arrow[ul, "(\xi_{\nu_1} \times \{r\}) \circ i_1"] \arrow[ur, "(\xi_{\nu_2} \times \{r\}) \circ i_2"'] &
\end{tikzcd}$$

Getting $(\xi_{\nu_1} \times \{0\}) \circ i_1 = (\xi_{\nu_2} \times \{0\}) \circ i_2$ *as maps from* N *to* W*, as the following diagram shows,*

$$\begin{tikzcd}
& W \arrow[dd, dashed, "\pi_W"] & \\
W_1 \arrow[ur, "\xi_{\nu_1} \times \{0\}"] & & W_2 \arrow[ul, "\xi_{\nu_2} \times \{0\}"'] \\
M \arrow[u, "i_M"] & N \arrow[ul, "i_1"] \arrow[ur, "i_2"'] \arrow[d, dashed] & P \arrow[u, "i_P"'] \\
& \emptyset &
\end{tikzcd}$$

We claim that

- W *is an oriented smooth manifold (see [Kos92], Chapter VI, Section 4).*
- W *is a* Γ*-proper space.* Pendiente
- W *is a* $\mathscr{H}_\Gamma^*$*-oriented space.* Pendiente

Proposition 4.14. *The relation in Definition 4.8 is an equivalence relation.*

Proof. 1. Reflexivity.

$$\begin{tikzcd}
& (M, \partial M) \arrow[dd, "\pi"] & \\
(M, \partial M) \arrow[ur, "Id"] \arrow[dr, "\pi"'] & & (M, \partial M) \arrow[ul, "Id"'] \arrow[dl, "\pi"] \\
& (X, A) &
\end{tikzcd}$$

2. Symmetry: Since the conditions for the relation $\sim$ have no laterally, it is a symmetric relation.

3. Transitivity: Consider the relations $(M, \partial M, \pi_M, x_M) \sim (N, \partial N, \pi_N, x_N)$ given by the maps $i_M : M \to W_1$ and $i_1 : N \to W_1$; and $(N, \partial N, \pi_N, x_N) \sim (P, \partial P, \pi_P, x_P)$ given by $i_P : P \to W_2$ and $i_2 : N \to W_2$. Consider the space W defined in Remark 4.13 for the maps $i_k : N \to W_k$, getting the following diagram,

$$
\begin{tikzcd}
 & W \arrow[dd, dashed, "\pi_W"] & \\
W_1 \arrow[ur, "\xi_{\nu_1}\times\{0\}"] & & W_2 \arrow[ul, "\xi_{\nu_2}\times\{0\}"'] \\
M \arrow[u, "i_M"] & N \arrow[ul, "i_1"'] \arrow[ur, "i_2"] & P \arrow[u, "i_P"']
\end{tikzcd}
$$

Since $i_M!(x_M) = i_1!(x_N)$ and $i_2!(x_N) = i_P!(x_P)$, it is satisfied the necessary condition that $((\xi_{\nu_1} \times \{0\}) \circ i_M)!(x_M) = ((\xi_{\nu_2} \times \{0\}) \circ i_P)!(x_P)$, by functoriality of pushforward. We define $\pi_W : W \to N$ by $\pi_W(d \oplus r) = \pi_{W_1}(\rho_1(d))$ if $d \in D_1$, and $\pi_W(d \oplus r) = \pi_{W_2}(\rho_2(d))$ if $d \in D_2$, as in the following diagram:

$$
\begin{tikzcd}
D_1 \arrow[d, "\rho_1"'] & D_1 \oplus \mathbb{R} \# D_2 \oplus \mathbb{R} \arrow[dd, dotted, "\pi_W"] & D_2 \arrow[d, "\rho_2"] \\
W_1 \arrow[ur, "\xi_{\nu_1}\times\{0\}"] \arrow[dr, "\pi_{W_1}"] & & W_2 \arrow[ul, "\xi_{\nu_2}\times\{0\}"'] \arrow[dl, "\pi_{W_2}"'] \\
M \arrow[u, "i_M"] \arrow[dr, "\pi_M"'] & N \arrow[ul, "i_1"'] \arrow[ur, "i_2"] \arrow[d, dotted] & P \arrow[u, "i_P"'] \arrow[dl, "\pi_P"] \\
 & X &
\end{tikzcd}
$$

Let us see that π_W is well defined, because

$$
\begin{aligned}
\pi_W((\xi_{\nu_1} \circ i_1(n)) \oplus r) &= \pi_{W_1}(\rho_1((\xi_{\nu_1} \circ i_1(n)) \oplus r)) \\
&= \pi_{W_1}(i_1(n)) \\
&= \pi_N(n)
\end{aligned}
$$

and $\pi_W((\xi_{\nu_2} \circ i_2(n)) \oplus -r) = \pi_N(n)$ too. Now, we must to verify that $\pi_M = \pi_W \circ \xi_{\nu_1} \circ i_M$: since $\pi_M = \pi_{W_1} \circ i_M$, we get

$$
\begin{aligned}
(\pi_W \circ \xi_{\nu_1} \circ i_M)(m) &= \pi_W(\xi_{\nu_1}(i_M(m)) \oplus 0) \\
&= \pi_{W_1}(\rho_1(\xi_{\nu_1}(i_M(m)))) \\
&= \pi_{W_1}((i_M(m))) \\
&= \pi_M(m);
\end{aligned}
$$

In a similar way we prove $\pi_P = \pi_W \circ \xi_{\nu_2} \circ i_P$. Then, we have proved that the cycles $(M, \partial M, \pi_M, x_M)$ and $(P, \partial P, \pi_P, x_P)$ are related by the cycle $(W, \partial W, \pi_W, (\xi_{\nu_1} \circ i_M)!(x_M))$, obtaining the transitivity of the desired equivalence pushforward relation.

$\square$

Remark 4.15. *If there is a proper smooth, Γ-equivariant, $\mathscr{H}_\Gamma$-orientation preserving map $f : (M, \partial M) \to (N, \partial N)$ such that $\pi_N \circ f = \pi_M$ and $f!(x_M) = x_N$, then*

$$[M, \partial M, \pi_M, x_M] = [N, \partial N, \pi_N, x_N],$$

i.e.,

$$[M, \partial M, \pi_N \circ f, x_M] = [N, \partial N, \pi_N, f!(x_M)].$$

The relation is obtained by the following commutative diagram

$$\begin{tikzcd}
 & N \arrow[d, dashed, "\pi_N"] & \\
M \arrow[ur, "f"] \arrow[dr, "\pi_M"'] & & N \arrow[ul, "Id"'] \arrow[dl, "\pi_N"] \\
 & X &
\end{tikzcd}$$

Definition 4.16. *For an $\mathscr{H}_\Gamma$-oriented manifold M with $\mathscr{H}_\Gamma$-orientation $\tau \in \mathscr{H}_\Gamma^{dim(M)}(TM)$, we define $-M$ as the same manifold M but with the $\mathscr{H}_\Gamma$-orientation $-\tau \in \mathscr{H}_\Gamma^{dim(M)}(TM)$.*

Remark 4.17. *Note that $(\mathscr{H}_*^{pf}(X, A; \Gamma), +)$ is actually a group:*

1. *Well defined: If we have*

$$[M_1, \partial M_1, \pi_{M_1}, x_{M_1}] = [M_2, \partial M_2, \pi_{M_2}, x_{M_2}]$$
$$[N_1, \partial N_1, \pi_{N_1}, x_{N_1}] = [N_2, \partial N_2, \pi_{N_2}, x_{N_2}],$$

where the cycles are related by the maps $f_i : M_i \to W_M$ and $g_i : N_i \to W_N$ for $i = 1, 2$, such that $f_1!(x_{M_1}) = f_2!(x_{M_2})$ and $g_1!(x_{N_1}) = g_2!(x_{N_2})$. With this information, we construct the diagram

$$\begin{tikzcd}
 & W_M \amalg W_N \arrow[d, dashed, "\pi_{W_M} \amalg \pi_{W_N}"] & \\
M_1 \amalg N_1 \arrow[ur, "f_1 \amalg g_1"] \arrow[dr, "\pi_{M_1} \amalg \pi_{N_1}"'] & & M_2 \amalg N_2 \arrow[ul, "f_2 \amalg g_2"'] \arrow[dl, "\pi_{M_2} \amalg \pi_{N_2}"] \\
 & X &
\end{tikzcd}$$

By Definition 4.8, in the cohomological elements we have that

$$\begin{aligned}
(f_1 \amalg g_1)!(x_{M_1} \oplus x_{N_1}) &= f_1!(x_{M_1}) \oplus g_1!(x_{N_1}) \\
&= f_2!(x_{M_2}) \oplus g_2!(x_{N_2}) \\
&= (f_2 \amalg g_2)!(x_{M_2} \oplus x_{N_2}),
\end{aligned}$$

and with this the group operation is well defined.

2. *Neutral element: We have that* $[M, \partial M, \pi_M, 0] \in \mathscr{H}_*^{pf}(X, A; \Gamma)$ *is the zero element, because for* $[N, \partial N, \pi_N, x_N] \in \mathscr{H}_*^{pf}(X, A; \Gamma)$*, we have that*

$$[N, \partial N, \pi_N, x_N] = [N \amalg M, \partial N \amalg \partial M, \pi_N \amalg \pi_M, x_N \oplus 0],$$

by the diagram

$$\begin{tikzcd}
 & N \amalg M \arrow[d, dashed, "\pi_N \amalg \pi_M"] & \\
N \arrow[ur, "i_1"] \arrow[dr, "\pi_N"'] & X & N \amalg M \arrow[ul, "Id"'] \arrow[dl, "\pi_N \amalg \pi_M"] \\
 & &
\end{tikzcd}$$

where $i_1!(x_N) = x_N \oplus 0$.

3. *Inverse element: Esta parte es suceptible al cambio de orientacion a compacta*
For a cycle $[M, \partial M, \pi_M, x_M] \in \mathscr{H}_*^{pf}(X, A; \Gamma)$ *the inverse element is* $[-M, -\partial M, \pi_M, x_M] \in \mathscr{H}_*^{pf}(X, A; \Gamma)$*, because their sum is equal to*

$$[M \amalg -M, \partial M \amalg -\partial M, \pi_M \amalg \pi_{-M}, x_M \oplus x_M].$$

The $\mathscr{H}_\Gamma^*$*-orientation of* M *provides us an* $\mathscr{H}_\Gamma^*$*-orientation of the tangent bundle* $\rho : TM \to M$*, which means that the zero section* $s : M \to TM$ *satisfy that* $s! = - \otimes \tau$*, where the class* $\tau \in \mathscr{H}_\Gamma^m(TM)$ *is such that the map* $- \otimes \tau : \mathscr{H}_\Gamma^*(M) \to \mathscr{H}_\Gamma^{*+m}(TM)$ *is an isomorphism. For* $-M$*, the zero section* $s^- : -M \to T(-M)$ *is the same* s *as a map, but we choose the opposite class satisfying that*

$$s^-! = - \otimes (-\tau) : \mathscr{H}_\Gamma^*(M) \to \mathscr{H}_\Gamma^{*+m}(TM).$$

Then, $s!(x_M) = x_M \otimes \tau = -(x_M \otimes -\tau) = -s^-!(x_M)$*. With this, we define the map* $s \amalg s^- : M \amalg -M \to TM$*, where*

$$(s \amalg s^-)! : \mathscr{H}_\Gamma^*(M \amalg -M) \cong \mathscr{H}_\Gamma^*(M) \oplus \mathscr{H}_\Gamma^*(-M) \to \mathscr{H}_\Gamma^{*+m}(TM)$$

satisfy that $(s \amalg s^-)!(x_M \oplus x_M) = 0 \in \mathscr{H}_\Gamma^*(TM)$*. Using the Remark 4.15 on the equality* $\pi_M \amalg \pi_{-M} = \pi_M \circ \rho \circ (s \amalg s^-)$*, we compute the cycle*

$$\begin{aligned}
0 &= [TM, \partial TM, \pi_M \circ \rho, (s \amalg s^-)!(x_M \oplus x_M)] \\
&= [M \amalg -M, \partial M \amalg -M, \pi_M \circ \rho \circ s \amalg s^-, x_M \oplus x_M] \\
&= [M \amalg -M, \partial M \amalg -M, \pi_M \amalg \pi_{-M}, x_M \oplus x_M].
\end{aligned}$$

Remark 4.18. *Esta parte es suceptible al cambio de orientacion a compacta*
If $[M, \partial M, \pi_M, x], [M, \partial M, \pi_M, y] \in \mathscr{H}_*^{pf}(X, A; \Gamma)$*, with* $x, y \in \mathscr{H}_\Gamma^{*+dim(M)}(M)$ *then*

$$[M, \partial M, \pi_M, x] + [M, \partial M, \pi_M, y] = [M, \partial M, \pi_M, x + y],$$

where $x + y \in \mathscr{H}_\Gamma^{*+dim(M)}(M)$.

Proof. We have to proof the following relation:

$$[M \amalg M, \partial M \amalg \partial M, \pi_M \amalg \pi_M, x \oplus y] = [M, \partial M, \pi_M, x + y]$$

where $x \oplus y \in \mathscr{H}^{*+dim(M)}(M \amalg M) \cong \mathscr{H}^{*+dim(M)}(M) \oplus \mathscr{H}^{*+dim(M)}(M)$. Similarly as in Remark 4.17, we consider $s \amalg s : M \amalg M \to TM$ in order to obtain

$$\begin{aligned}(s \amalg s)!(x \oplus y) &= (s \amalg s)!(x \oplus 0) + (s \amalg s)!(0 \oplus y) = s!(x) + s!(y) \\ &= x \otimes \tau + y \otimes \tau = (x + y) \otimes \tau = s!(x + y),\end{aligned}$$

and with this, in cycles we get that

$$\begin{aligned}[M \amalg M, \partial M \amalg \partial M, \pi_M \amalg \pi_M, x \oplus y] &= [M \amalg M, \partial M \amalg M, \pi_M \circ \rho \circ (s \amalg s), x \oplus y] \\ &= [TM, \partial TM, \pi_M \circ \rho, (s \amalg s)!(x \oplus y)] \\ &= [TM, \partial TM, \pi_M \circ \rho, s!(x + y)] \\ &= [M, \partial M, \pi_M \circ \rho \circ s, x + y] \\ &= [M, \partial M, \pi_M, x + y],\end{aligned}$$

as we desired, where the second and fourth equalities are given by Remark 4.15 applied to $s \amalg s$ and s respectively. □

Remark 4.19. *Thom: if $\rho_E : E \to M$ is a Γ-vector bundle, then $(M, \partial M, \pi_M, x_M) \sim (S(E), S(E)|_{\partial M}, \pi_M \circ \rho_{S(E)})$ Redactar mejor y demostrar.*

Revisar con Mario If $\rho_E : E \to M$ is a Γ-vector bundle, then $(M, \partial M, \pi_M, x_M) \sim (D(E), D(E)|_{\partial M}, \pi_M \circ \rho_E|_{D(E)}, i^*_{D(E)}(\xi_E!(x_M)))$, such as the following diagram suggest

$$\begin{tikzcd} M \arrow[r, "\xi_E"] \arrow[dr, "Id_M"'] & E \arrow[d, "\rho_E"] & D(E) \arrow[l, "i_{D(E)}"'] \arrow[dl, "\rho_E|_{D(E)}"] \\ & M & \end{tikzcd}$$

where $i^*_{D(E)} \circ \xi_E!(x_M) = \xi_E!(x_M)$ by the fact that $Im(\xi_E) \subseteq D(E)$.

Definition 4.20. *For a map $g : (X, A) \to (Y, B)$, define $\mathscr{H}^{pf}_*(g) := g_*$ by*

$$g_* : \mathscr{H}^{pf}_*(X, A; \Gamma) \to \mathscr{H}^{pf}_*(Y, B; \Gamma) : [M, \partial M, \pi, x] \mapsto [M, \partial M, g \circ \pi, x].$$

Proposition 4.21. *The pushforward homology is functorial.*

Proof. We have that $(h \circ g)_* = h_* \circ g_*$ and it is well defined because, if $[M, \partial M, \pi, x] = [N, \partial N, \pi', x'] \in \mathscr{H}^{pf}_*(X, A; \Gamma)$, i.e., there exist a cycle $[W, \partial W, \widehat{\pi}, \widehat{x}]$ over (X, A) and maps $f : (M, \partial M) \to (W, \partial W)$ and $f' : (N, \partial N) \to (W, \partial W)$ such that

$$\begin{aligned}\pi &= \widehat{\pi} \circ f \\ \pi' &= \widehat{\pi} \circ f' \\ f!(x) &= \widehat{x} = f'!(x'),\end{aligned}$$

then composing with g we get

$$\begin{aligned} g \circ \pi &= (g \circ \widehat{\pi}) \circ f \\ g \circ \pi' &= (g \circ \widehat{\pi}) \circ f' \\ f!(x) &= \widehat{x} = f'!(x'), \end{aligned}$$

and we get that $[M, \partial M, g \circ \pi, x] = [N, \partial N, g \circ \pi', x'] \in \mathscr{H}_*^{pf}(Y, B; \Gamma)$, where $g \circ \pi$ and $g \circ \pi'$ are proper Γ-equivariant maps of pairs too by composition, showing that g_* is well defined.

□

Now, we want to prove that the groups $\mathscr{H}_*^{pf}$ define a Γ-equivariant homology theory over the cocompact proper Γ-CW pairs, showing each of the axioms that it has to satisfy (cf. Definition 4.1). This is the main objective of the thesis, and in this work we will spend the rest of the chapter.

Remark 4.22. *When we have an homotopy between CW-spaces maps $X \to Y$, this is regularly identified by a map $[0,1] \times X \to Y$ (cf. [GP10], p.33), but when we deal with homotopies between smooth manifolds $M \to N$, we identify this homotopy with a smooth map $\mathbb{R} \times M \to N$, in order to avoid problems with the boundary and corners.*

Proposition 4.23. *The pushforward homology satisfies Γ-homotopy invariance, i.e., if $g_0, g_1 : (X, A) \to (Y, B)$ are proper Γ-homotopic maps of cocompact proper Γ-CW pairs, then for all $n \in \mathbb{Z}$, the induced maps are equal*

$$g_{0*} = g_{1*} : \mathscr{H}_n^{pf}(X, A; \Gamma) \to \mathscr{H}_n^{pf}(Y, B; \Gamma)$$

Proof. Given a proper Γ-homotopy $H : \mathbb{R} \times (X, A) \to (Y, B)$ between g_0 and g_1, we have to show that $g_{0*} = g_{1*}$, i.e., for a cycle $[M, \partial M, \pi, x] \in \mathscr{H}_n^{pf}(X, A; \Gamma)$ we need that

$$[M, \partial M, g_0 \circ \pi, x] = [M, \partial M, g_1 \circ \pi, x] \in \mathscr{H}_n^{pf}(Y, B; \Gamma).$$

To prove this, consider the tuple $(\mathbb{R} \times M, \mathbb{R} \times \partial M, \widehat{\pi}, x)$ and maps $f_0, f_1 : M \to \mathbb{R} \times M$, where

$$\widehat{\pi} = H \circ (id_{\mathbb{R}} \times \pi) : (\mathbb{R} \times M, \mathbb{R} \times \partial M) \to (Y, B)$$

and $f_i(m) = (i, m)$ for $i = 0, 1$ and all $m \in M$.

$$\begin{tikzcd}
 & (\mathbb{R} \times M, \mathbb{R} \times \partial M) \arrow[d, "Id_{\mathbb{R}} \times \pi"] & \\
(M, \partial M) \arrow[ur, "f_0"] \arrow[dr, hook, "g_0 \circ \pi"'] & \mathbb{R} \times (X, A) \arrow[d, "H"] & (M, \partial M) \arrow[ul, "f_1"'] \arrow[dl, hook, "g_1 \circ \pi"] \\
 & (Y, B) &
\end{tikzcd}$$

In this case, we get that $\widehat{\pi}\circ f_i = g_i\circ\pi$ for $i=0,1$, and for the cohomology element $x\in\mathscr{H}_\Gamma^{n+dim(M)}(M)$, we get that $f_0!(x)=f_1!(x)$ in $\mathscr{H}_\Gamma^{n+dim(M)+1}(\mathbb{R}\times M)$, because considering the proyection $\rho:\mathbb{R}\times M\to M$, then

$$(\rho\circ f_1)! = (\rho\circ f_2)!$$

where by the pushforward structure, $\rho!$ is an isomorphism, and by functoriality $f_1!(x)=f_2!(x)$. □

In order to prove the long exact sequence condition, previously we discuss the following remark.

Remark 4.24. *If we have that $[M,\partial M,\pi_M,x_M]=[N,\partial N,\pi_N,x_N]$ by the cycle $[W,\partial W,\pi_W,x_W]$, then by the Remark 4.15, $[M,\partial M,\pi_M,x_M]=[W,\partial W,\pi_W,x_W]$ in $\mathscr{H}_*^{pf}(X,A;\Gamma)$, with the map $f:M\to W$ satisfying that $\pi_W\circ f=\pi_M$ and $f!(x_M)=x_W$.*

Proposition 4.25. *The pushforward homology satisfies the long exact sequence axiom.*

Proof. For the sequence of cocompact proper Γ-CW pairs

$$(A,\emptyset)\xrightarrow{\ j\ }(X,\emptyset)\xrightarrow{\ i\ }(X,A)$$

We have to prove that the sequence

$$\mathscr{H}_*^{pf}(A,\emptyset;\Gamma)\xrightarrow{\ j_*\ }\mathscr{H}_*^{pf}(X,\emptyset;\Gamma)\xrightarrow{\ i_*\ }\mathscr{H}_*^{pf}(X,A;\Gamma)\xrightarrow{\ \partial\ }\mathscr{H}_{*-1}^{pf}(A,\emptyset;\Gamma)$$

is an exact sequence, were the maps are defined as

- $j_*([M,\emptyset,\pi_M,x_M])=[M,\emptyset,j\circ\pi_M,x_M]$.
- $i_*([M,\emptyset,\pi_M,x_M])=[M,\emptyset,\pi_M,x_M]$.
- $\partial([M,\partial M,\pi_M,x_M])=[\partial M,\emptyset,\pi_M|_{\partial M},r_\partial(x_M)]$, where $r_\partial=i_{\partial M}^*$ with $i_{\partial M}:\partial M\hookrightarrow M$ the canonical inclusion. This map is well defined, because if $[M,\partial M,\pi_M,x_M]=[N,\partial N,\pi_N,x_N]\in\mathscr{H}_*^{pf}(X,A;\Gamma)$ by the cycle $(W,\partial W,\pi_W,x_W)$ where $f!(x_M)=x_W=g!(x_N)$. From the restriction diagram,

$$\begin{tikzcd} M \arrow[r, "f"] & W & N \arrow[l, "g"'] \\ \partial M \arrow[u, "i_{\partial M}"] \arrow[r, "f|_{\partial M}"'] & \partial W \arrow[u, "i_{\partial W}"'] & \partial N \arrow[u, "i_{\partial N}"'] \arrow[l, "g|_{\partial N}"] \end{tikzcd}$$

by Definition 4.8, we get the commutative diagram

$$\begin{tikzcd} \mathscr{H}_\Gamma^*(M) \arrow[r, "f!"] \arrow[d, "i_{\partial M}^*"'] & \mathscr{H}_\Gamma^*(W) \arrow[d, "i_{\partial W}^*"] & \mathscr{H}_\Gamma^*(N) \arrow[l, "g!"'] \arrow[d, "i_{\partial N}^*"] \\ \mathscr{H}_\Gamma^*(\partial M) \arrow[r, "f|_{\partial M}!"'] & \mathscr{H}_\Gamma^*(\partial W) & \mathscr{H}_\Gamma^*(\partial N) \arrow[l, "g|_{\partial N}!"] \end{tikzcd}$$

Then, we get that $[\partial M, \emptyset, \pi_M|_{\partial M}, r_\partial(x_M)] = [\partial N, \emptyset, \pi_N|_{\partial N}, r_\partial(x_N)]$ in $\mathscr{H}_*^{pf}(A, \emptyset; \Gamma)$, by the cycle $(\partial W, \emptyset, \pi_W|_{\partial W}, i^*_{\partial W}(x_W))$,

$$\begin{aligned} f|_{\partial M}!(i^*_{\partial M}(x_M)) &= i^*_{\partial W}(f!(x_M)) \\ &= i^*_{\partial W}(g!(x_N)) \\ &= g|_{\partial N}!(i^*_{\partial N}(x_N)). \end{aligned}$$

Exactness in $\mathscr{H}_*^{pf}(X, \emptyset; \Gamma)$:

- $Ker(i_*) \subseteq Im(j_*)$: Let $[M, \emptyset, \pi_M, x_M] \in Ker(i_*)$, i.e., it has a related cycle $[N, \partial N, \pi_N, 0] \in \mathscr{H}_*^{pf}(X, A; \Gamma)$ by a Γ-proper manifold pair $(W, \partial W)$, with diagram

$$\begin{tikzcd} & W \arrow[dd, dashed, "\pi_W"] & \\ M \arrow[ur, "f_M"] \arrow[dr, "\pi_M"'] & & N \arrow[ul, "f_N"'] \arrow[dl, "\pi_N"] \\ & X & \end{tikzcd}$$

such that $f_M!(x_M) = f_N!(0) = 0 \in \mathscr{H}_\Gamma^*(W)$. Since $f_M : (M, \emptyset) \to (W, \partial W)$ can be factorized by

$$M \xrightarrow{f_M^0} W^0 \xrightarrow{i_{W^0}} W$$

and with this, $f_M^0!(x_M) \in Ker(i_{W^0}!)$, by functoriality of the pushforward. Considering the long exact sequence in the cohomology theory for the pair $(W, \partial W)$, we get

$$\mathscr{H}_\Gamma^{*-1}(\partial W) \xrightarrow{\partial} \mathscr{H}_\Gamma^*(W^0) \xrightarrow{i_{W^0}!} \mathscr{H}_\Gamma^*(W)$$

By exactness, we have that there exist $y \in \mathscr{H}_\Gamma^{*-1}(\partial W)$ such that $\partial(y) = f_M!(x_M)$. Consider the inclusion through a collar $l : \partial W \to W^0$, and therefore by the definition of pushforward structure, $l! : \mathscr{H}_\Gamma^{*-1}(\partial W) \to \mathscr{H}_\Gamma^*(\partial W^0)$ is identified with the cohomological map ∂. With this, we claim that $j_*([\partial W, \emptyset, \pi_W|_{\partial W}, y]) = [M, \emptyset, \pi_M, x_M]$:

$$\begin{aligned} j_*([\partial W, \emptyset, \pi_W|_{\partial W}, y]) &= [\partial W, \emptyset, j \circ \pi_W|_{\partial W}, y] \\ &= [\partial W, \emptyset, \pi_W \circ l, y] \\ &= [W^0, \emptyset, \pi_W|_{W^0}, l!(y)] \\ &= [W^0, \emptyset, \pi_W|_{W^0}, \partial(y)] \\ &= [W^0, \emptyset, \pi_W|_{W^0}, f_M^0!(x_M)] \\ &= [M, \emptyset, \pi_W \circ f_M, x_M] \\ &= [M, \emptyset, \pi_M, x_M], \end{aligned}$$

where: the first one is the definition of j_*; second is by the fact that $j \circ \pi_W|_{\partial W}$ is homotopic to $\pi_W \circ l$ by the the homotopy between $i_{\partial W} : \partial W \to W$ and $l : \partial W \to W^0 \subset W$; third one is by Remark 4.15 aplied to the map $l : W^0 \hookrightarrow W$; fourth is by $l! = \partial$; fifth is $\partial(y) = f_M!(x_M)$; sixth is Remark 4.15 applied to f_M^0 and the fact that $\pi_W|_{W^0} \circ f_M^0 = \pi_W \circ f_M$; and the last one is $\pi_M = \pi_W \circ f_M$.

- $Ker(i_*) \supseteq Im(j_*)$: We have to show that for a cycle $[M, \emptyset, \pi_M, x_M] \in \mathscr{H}_*^{pf}(A, \emptyset; \Gamma)$, we have that $i_*(j_*([M, \emptyset, \pi_M, x_M])) = 0$. We have that

$$i_*(j_*([M, \emptyset, \pi_M, x_M])) = [M, \emptyset, j \circ \pi_M, x_M] \in \mathscr{H}_*^{pf}(X, A; \Gamma).$$

We have that this cycle is related with the zero cycle

$$[M \times [0,1], M \times \{1\}, \pi_M \circ \pi_1, 0]$$

by the commutative diagram

$$\begin{tikzcd}
(M, \emptyset) \arrow[rr, "Id \times 0"] \arrow[dr, "j \circ \pi_M"'] & & (M \times [0,1], M \times \{1\}) \arrow[dl, "j \circ \pi_M \circ \pi_1"] \\
& (X, A) &
\end{tikzcd}$$

where we have that $(Id_M \times 0)!(x_M) = 0$ because

$$\mathscr{H}_\Gamma^*(M \times [0,1], M \times \{1\}) = \{0\},$$

by the fact that $i_{M \times \{1\}} : M \times \{1\} \to M \times [0,1]$ is a Γ-homotopy equivalence, therefore in the long exact sequence $i^n_{M \times \{1\}}$ is an isomorphism for all $n \in \mathbb{N}$:

$$\begin{tikzcd}
\cdots \arrow[r] & \mathscr{H}_\Gamma^{*-1}(M \times [0,1]) \arrow[r, "i^{*-1}_{M \times \{1\}}"] & \mathscr{H}_\Gamma^{*-1}(M \times \{1\}) \arrow[dl] \\
& \mathscr{H}_\Gamma^*(M \times [0,1], M \times \{1\}) \arrow[dl] & \\
\mathscr{H}_\Gamma^*(M \times [0,1]) \arrow[r, "i^*_{M \times \{1\}}"] & \mathscr{H}_\Gamma^*(M \times \{1\}) \arrow[r] & \cdots
\end{tikzcd}$$

Exactness in $\mathscr{H}_*^{pf}(X, A; \Gamma)$:

- $Ker(\partial) \supseteq Im(i_*)$: For $[M, \emptyset, \pi_M, x_M] \in \mathscr{H}_*^{pf}(X, \emptyset; \Gamma)$, we have that

$$\partial(i_*([M, \emptyset, \pi_M, x_M])) = \partial([M, \emptyset, \pi_M, x_M]) = [\emptyset, \partial\emptyset, \pi_M|_\emptyset, r_\partial(x_M)] = 0,$$

considering that $\mathscr{H}_\Gamma^*(\emptyset) = \{0\}$.

- $Ker(\partial) \subseteq Im(i_*)$: Let $[M, \partial M, \pi_M, x_M] \in Ker(\partial) \subseteq \mathscr{H}_*^{pf}(X, A; \Gamma)$, i.e., that $[\partial M, \emptyset, \pi_M|_{\partial M}, r_\partial(x_M)] = 0 \in \mathscr{H}_*^{pf}(A, \emptyset; \Gamma)$. Then, $[\partial M, \emptyset, \pi_M|_{\partial M}, r_\partial(x_M)] = [W, \emptyset, \pi_W, 0]$ by the existence of a commutative diagram

$$\begin{tikzcd} \partial M \arrow[rr, "f"] \arrow[dr, "\pi_M|_{\partial M}"'] & & W \arrow[dl, "\pi_W"] \\ & A & \end{tikzcd}$$

such that $f!(r_\partial(x_M)) = 0$ and W a manifold without boundary. By Lemma 4.10, we can consider f a Γ-embedding, and then applying Lemma 4.12 to $i_{M^0} \circ l : \partial M \to M$ and $f : \partial M \to W$, to get that there exist Γ-vector bundles $\rho_{D_M} : D_M \to M$ and $\rho_{D_W} : D_W \to W$ such that $\nu_M := \nu_{\xi_{D_M} \circ i_{M^0} \circ l} \cong \nu_{\xi_{D_W} \circ f} =: \nu_W$, and Remark 4.13 in order to define

$$T := (D_M \oplus \mathbb{R})_{\nu_M \oplus \mathbb{R}} \#_{\nu_W \oplus \mathbb{R}} (D_W \oplus \mathbb{R}),$$

We claim that

1. ∂T is the identification of $\partial(D_M) \cong D_M|_{\partial M}$ pendiente

We have $\xi_{D_M \oplus \mathbb{R}} \circ i_{M^0} \circ l : \partial M \to D_M$ and $\xi_{D_W \oplus \mathbb{R}} \circ f : \partial M \to D_W$ are equals in T, i.e., composing with $i_{D_M \oplus \mathbb{R}}$ and $i_{D_W \oplus \mathbb{R}}$ respectively by Remark 4.13.

With this, defining $\iota : M \to T$ as the natural embedding $i_{D_M \times \{0\}} \circ \xi_{D_M \oplus \mathbb{R}}$, and $\pi_T : T \to X$ is defined by $\pi_T(a) = \pi_M \circ \rho_{D_M \oplus \mathbb{R}}$ if $a \in D_M \oplus \mathbb{R}$, or $\pi_T(a) = \pi_W \circ \rho_{D_W \oplus \mathbb{R}}$ if $a \in D_M \oplus \mathbb{R}$, such as in Proposition 4.14 obtaining in the same way that $\pi_T \circ \iota = \pi_M$. We get that $\iota \circ i_{\partial M} = i_{D_W \times \{0\}} \circ \xi_{D_W \oplus \mathbb{R}} \circ f$, where identifying $i_{\partial T} = i_{D_W \times \{0\}}$, obtaining the commutative diagram

$$\begin{tikzcd} \partial M \arrow[r, "i_{\partial M}"] \arrow[d, "\xi_{D_W \oplus \mathbb{R}} \circ f"'] & M \arrow[d, "\iota"] \\ \partial T \arrow[r, "i_{\partial T}"'] & T \end{tikzcd}$$

therefore by Definition 4.8

$$i_{\partial T}^* \circ \iota!(x_M) = (\xi_{D_W \oplus \mathbb{R}} \circ f)! \circ i_{\partial M}^*(x_M) = 0$$

Since $\iota!(x_M) \in Ker(r_\partial)$, by the long exact sequence in cohomology for the pair $(T, \partial T)$, there exist $y \in \mathscr{H}_\Gamma^*(T^0)$ such that $i_{T^0}!(y) = \iota!(x_M)$, where

$i_{T^0} : T^0 \hookrightarrow T$. Then

$$\begin{aligned} i_*([T^0, \emptyset, \pi_T|_{T^0}, y]) &= [T^0, \emptyset, \pi_T|_{T^0}, y] \\ &= [T^0, \emptyset, \pi_T \circ i_{T^0}, y] \\ &= [T, \partial T, \pi_T, i_{T^0}!(y)] \\ &= [T, \partial T, \pi_T, \iota!(x_M)] \\ &= [M, \partial M, \pi_T \circ \iota, x_M] \\ &= [M, \partial M, \pi_M, x_M] \end{aligned}$$

where: first equality is definition of i_*; second is a direct composition; third is By Remark 4.15 on i_{T^0}; fourth is the obtained pushforward equality; fifth is by Remark 4.15 on ι; and sixth is the equality $\pi_M = \pi_T \circ \iota$.

Exactness in $\mathscr{H}_*^{pf}(A, \emptyset; \Gamma)$:

- $Ker(j_{*-1}) \supseteq Im(\partial)$: for $[M, \partial M, \pi_M, x_M] \in \mathscr{H}_*^{pf}(X, A; \Gamma)$, we note that $j_{*-1}(\partial([M, \partial M, \pi_M, x_M])) = [\partial M, \emptyset, j \circ (\pi_M|_{\partial M}), r_\partial(x_M)] \in \mathscr{H}_*^{pf}(X, \emptyset; \Gamma)$ is the zero class, because by the cohomology long exact sequence

$$\mathscr{H}_\Gamma^*(M^0) \xrightarrow{i^0! = j_M^*} \mathscr{H}_\Gamma^*(M) \xrightarrow{i_{\partial M}^*} \mathscr{H}_\Gamma^*(\partial M) \xrightarrow{l! = \partial} \mathscr{H}_\Gamma^{*+1}(M^0)$$

  we have that $l!(r_\partial(x_M)) = 0$, then

$$\begin{aligned} [\partial M, \emptyset, j \circ (\pi_M|_{\partial M}), r_\partial(x_M)] &= [\partial M, \emptyset, j \circ \pi_M \circ l, r_\partial(x_M)] \\ &= [M^0, \emptyset, j \circ \pi_M, l!(r_\partial(x_M))] \\ &= [M^0, \emptyset, j \circ \pi_M, 0] = 0, \end{aligned}$$

  where the first equation is by the homotopy given along the collar and the second is Remark 4.15 on $l : \partial M \to M^0$.

- $Ker(j_{*-1}) \subseteq Im(\partial)$: If $[M, \emptyset, \pi_M, x_M] \in ker(j_*) \subseteq \mathscr{H}_*^{pf}(A, \emptyset; \Gamma)$, then $j_*([M, \emptyset, \pi_M, x_M]) = [M, \emptyset, j \circ \pi_M, x_M] = 0$, i.e., by Remark 4.24 there exist a description of the null element $[W, \emptyset, \pi_W, 0_W] \in \mathscr{H}_*^{pf}(X, \emptyset; \Gamma)$ with a map $f : M \to W$ such that $\pi_N \circ f = \pi_M$ and $f!(x_M) = 0_W$.

  Applying the Theorem 2.5 and denoting $V = \nu_M^E$, we get

$$\begin{tikzcd} V \arrow[r, "\eta_f"] \arrow[d] & E \arrow[d] \\ M \arrow[r, "f"] & W \end{tikzcd}$$

  Since $f = \rho_E \circ \eta_f \circ \xi_V$, we have that $\eta_f!(\xi_V!(x_M)) = \xi_E!(f!(x_M)) = 0_E$. Consider

$$T := DNC(E, M)|_{[0,1)} = V \times \{0\} \sqcup E \times (0, 1)$$

where $\partial T = V \times \{0\}$. Using the map $l : \partial T \to T^0$ homotopic to the tubular neighborhood embedding, since l factors through η_f up to homotopy, we have that $l!(\xi_V!(x_M)) = 0_{T^0}$.

Consider the cohomology long exact sequence of the pair $(T, \partial T)$,

$$\mathscr{H}^*_\Gamma(T^0) \xrightarrow{i_{T^0}!=j_T^*} \mathscr{H}^*_\Gamma(T) \xrightarrow{r_\partial = i^*_{\partial T}} \mathscr{H}^*_\Gamma(\partial T) \xrightarrow{l!=\partial} \mathscr{H}^{*+1}_\Gamma(T^0)$$

and since $\mathscr{H}^*_\Gamma(\partial T) \cong \mathscr{H}^*_\Gamma(M)$ by the Thom isomorphism given by the zero section $\xi_V : M \to V$ in the pushforward structure (see Definition 4.7), we get that there exists $y \in \mathscr{H}^*_\Gamma(T)$ such that $r_\partial(y) = \xi_V!(x_M)$. With this we get as we need that $\partial([T, \partial T, \pi_T, y]) = [M, \emptyset, \pi_M, x_M]$, where we define $\pi_T : T \to X$ by

$$\pi_T((x,t)) = \begin{array}{ll} \pi_W \circ \rho_E(x) & \text{if } t \neq 0 \\ \pi_M \circ \rho_V(x) & \text{if } t = 0. \end{array}$$

We show this recalling that $\pi_T|_{\partial T} = \pi_M \circ \rho_V$, and with this

$$\begin{aligned} \partial([T, \partial T, \pi_T, y]) &= [\partial T, \emptyset, \pi_{\partial T}, r_\partial(y)] \\ &= [V, \emptyset, \pi_M \circ \rho_V, \xi_V!(x_M)] \\ &= [M, \emptyset, \pi_M \circ \rho_V \circ \xi_V, x_M] \\ &= [M, \emptyset, \pi_M, x_M], \end{aligned}$$

where the first equality is by definition of ∂; second equality is by $\partial T \cong V$, definition of $\pi_{\partial T} = \pi_T|_{\partial T}$ and $r_\partial(y) = \xi_V!(x_M)$; third equation is by Remark 4.14 applied to ξ_V; and last equation is by ξ_V is a section of $\rho_V : V \to M$.

□

Proposition 4.26. *The pushforward homology satisfy the excision property. i.e., for a map $\phi : (X, A) \to (Y, B)$ such that $Y \cong X \cup_A B$, we get that $\mathscr{H}^{pf}_*(X, A) \cong \mathscr{H}^{pf}_*(X \cup_A B, B)$.*

Proof. First, suppose that ϕ is an equivariant cellular inclusion. We will show that ϕ_* is an isomorphism. In this case the inverse for the map

$$[M, \partial M, \pi_M, x_M] \in \mathscr{H}^{pf}_*(X, A) \xmapsto{\phi_*} [M, \partial M, \phi \circ \pi_M, x_M] \in \mathscr{H}^{pf}_*(Y, B)$$

can be described as follows: for $[M, \partial M, \pi_M, x_M] \in \mathscr{H}^{pf}_*(Y, B)$, consider $U \subseteq Y$ an open set such that there exist a strongly deformation retract $r : U \to X$ (cf. [LO01], pg. 587); with this, consider the open submanifold $M_U = \pi_M^{-1}(U)$ of M, and then using the inclusion $i_{M_U} : M_U \to M$ define $h : \mathscr{H}^{pf}_*(Y, B) \to \mathscr{H}^{pf}_*(X, A)$ by

$$h([M, \partial M, \pi_M, x_M]) = [M_U, \partial M_U, r \circ \pi_M|_{M_U}, i^*_{M_U} x_M].$$

We claim that $\phi_*^{-1} = h$.

- $h \circ \phi_* = Id_{\mathscr{H}_*^{pf}(X,A;\Gamma)}$:

$$\begin{tikzcd}
M \arrow[d, "\pi_M"] & M_U = (\phi \circ \pi_M)^{-1}(U) \arrow[l, hook', "i_{M_U}"'] \arrow[d, "\pi_M|_{M_U}"] & \\
X \arrow[r, hook, "i_X"] \arrow[dr, hook, "\phi"'] & U \arrow[r, "r"] \arrow[d, "i_U"] & X \\
& Y &
\end{tikzcd}$$

Since $X \subseteq U$, we get that $(\phi \circ \pi_M)^{-1}(U) \supseteq \pi_M^{-1}(X) = M$, therefore $M_U = M$, and $i_{M_U} = id_M$, then

$$\begin{aligned}
h \circ \phi_*([M, \partial M, \pi_M, x_M]) &= h([M, \partial M, \phi \circ \pi_M, x_M]) \\
&= [M_U, \partial M_U, r \circ \phi \circ \pi_M|_{M_U}, i_{M_U}^*(x_M)] \\
&= [M, \partial M, r \circ \phi \circ \pi_M, x_M] \\
&= [M, \partial M, \pi_M, x_M],
\end{aligned}$$

where first and second equations are obtained by definitions of ϕ_* and h, respectively; third equation is by $i_{M_U} = Id_M$ and $Id_M^* = Id_{\mathscr{H}_\Gamma^*(M)}$; and fourth is get by the fact that $r \circ \phi = r|_X \cong_h Id_X$.

- $\phi_* \circ h = Id_{\mathscr{H}_*^{pf}(Y,B;\Gamma)}$:

$$\begin{tikzcd}
M \arrow[r, "\pi_M"'] & Y & \\
M_U \arrow[u, hook, "i_{M_U}"] \arrow[r, "\pi_M|_{M_U}"'] & U \arrow[u, hook, "i_U"] \arrow[r, "r"'] & X \arrow[ul, hook, "\phi"']
\end{tikzcd}$$

$$\begin{aligned}
\phi_* \circ h([M, \partial M, \pi_M, x_M]) &= \phi_*([M_U, \partial M_U, r \circ \pi_M|_{M_U}, i_{M_U}^* x_M]) \\
&= [M_U, \partial M_U, \phi \circ r \circ \pi_M|_{M_U}, i_{M_U}^* x_M].
\end{aligned}$$

Consider $\widetilde{W} = M \times (0,1] \cup M_U \times \{0\}$, and with this, define

$$W = \widetilde{W} \cup_{\partial M \times (0,1] \cup \partial M_U \times \{0\}} \widetilde{W}.$$

Denote by $\lambda : W \hookrightarrow (M \times [0,1]) \cup_{\partial M \times [0,1]} (M \times [0,1])$ the inclusion map, and denote

$$i_0 : -M_U \to W^0, \qquad i_1 : M \to W^0$$

the inclusions given by a collar.

W is a proper $\mathscr{H}_\Gamma$-oriented manifold with boundary, where

$$-M_U \times \{0\} \bigcup M \times \{1\} \subseteq \partial W,$$

i.e., there exists an embedding $i! = (i_0 \sqcup i_1)! : \mathscr{H}^*(M_U) \oplus \mathscr{H}^*(M) \to \mathscr{H}^{*+1}(W^0)$. Consider the long exact sequence of the pair $(W, \partial W)$,

$$\cdots \longrightarrow \mathscr{H}^*_\Gamma(W) \xhookrightarrow{r_\partial} \mathscr{H}^*_\Gamma(\partial W) \xrightarrow{i!} \mathscr{H}^{*+1}_\Gamma(W^0) \longrightarrow \cdots$$

We want to see that $i!((x_M, -i^*_{M_U}(x_M))) = (i_0 \sqcup i_1)!(x_M - i^*_{M_U}(x_M)) = 0$, which is equivalent to the fact $x_M - i^*_{M_U}(x_M) \in Im(r_\partial)$. To construct a desired cycle, consider the cohomology element obtained by gluing[4] two copies of x_M along $\partial M \times [0,1]$

$$\hat{x} := \left(x_M \bigcup_{\partial M \times [0,1]} x_M \right) \in \mathscr{H}^*_\Gamma\left(M \times [0,1] \bigcup_{\partial M \times [0,1]} M \times [0,1] \right) \cong \mathscr{H}^*_\Gamma(D(M)).$$

With this, we get that $r_\partial(\lambda^*(\hat{x})) = x_M - i^*_{M_U}(x_M)$. It implies $(i_1)!(x_M) = (i_0)!(i^*_{M_U}(x_M))$, and with this,

$$\begin{aligned}
[M_U, \partial M_U, \phi \circ r \circ \pi_M|_{M_U}, i^*_{M_U} x_M] &= [M_U, \partial M_U, \phi \circ r \circ \pi_M \circ i_{M_U}, i^*_{M_U} x_M] \\
&= [M_U, \partial M_U, \pi_{W^0} \circ i_0, i^*_{M_U} x_M] \\
&= [W^0, \emptyset, \pi_{W^0}, i_0!(i^*_{M_U} x_M)] \\
&= [W^0, \emptyset, \pi_{W^0}, i_1!(x_M)] \\
&= [M, \partial M, \pi_{W^0} \circ i_1, (x_M)] \\
&= [M, \partial M, \pi_M, x_M],
\end{aligned}$$

where the first equality is by definition of restriction; second equality is given by the fact that $\phi \circ r \circ \pi_M \circ i_{M_U}$ is Γ-homotopic to $\pi_{W^0} \circ i_0$; third equality is by Remark 4.15 applied to $i_0 : M_U \to W^0$; fourth equation is given by $(i_1)!(x_M) = (i_0)!(i^*_{M_U}(x_M))$; fifth equation is by Remark 4.15 applied to $i_1 : M \to W^0$; and last equation is given by $\pi_M = \pi^0_W \circ i_1$.

For general $\phi : (X, A) \to (Y, B)$, we could change (Y, B) by the mapping cylinder $(M_\phi, M_{\phi|A})$ and use excision for inclusions (because there is an homotopy equivalence $\psi : (Y, B) \to (M_\phi, M_{\phi|A})$ and therefore, its cohomology groups are isomorphic). □

Proposition 4.27. *(Opcion 2) The pushforward homology satisfy the excision property, i.e., for a proper Γ-CW pair (X, A), there exist an isomorphism*

$$\mathscr{H}^{pf}_*(X, A) \cong \mathscr{H}^{pf}_*(X \setminus A)$$

Proof. Consider the map

$$h : \mathscr{H}^{pf}_*(X, A; \Gamma) \to \mathscr{H}^{pf}_*(X \setminus A, \emptyset; \Gamma)$$

[4] In this case $\hat{x}$ can be obtained in the Mayer-Vietoris cohomology sequence as the element in the cohomology of the double of M, denoted by $D(M)$, such that its image under the restriction map hold $r_{M \times [0,1]}(\hat{x}) = (x_M, x_M)$.

defined in the following way: for $[M, \partial M, \pi_M, x_M] \in \mathscr{H}_*^{pf}(X, A; \Gamma)$, consider an open set $U \subseteq X$ such that $\pi_M^{-1}(A) \subseteq \pi_M^{-1}(U)$, where $\pi_M^{-1}(U)$ is an open set in M. There exist an open set $\tilde{U} \subseteq M$ such that there is a strong deformation retract $\alpha : \tilde{U} \times [0,1] \to \tilde{U}$ such that $\alpha_0 = Id_{\tilde{U}}$, and $\alpha_1 : \tilde{U} \to C \subseteq \tilde{U}$ such that $\alpha_1|_C = Id_C$. There exist a closed $C \subseteq M$ such that $\pi_M^{-1}(U) \subseteq C \subseteq \tilde{U}$. Therefore we have the succesive contenations

$$\pi_M^{-1}(A) \subset \pi_M^{-1}(U) \subset C \subset \tilde{U} \subset M$$

With this situation we get that

1. $x_M \in \mathscr{H}_\Gamma^*(M)$.
2. $x_M|_C = (i_C^M)^*(x_M) \in \mathscr{H}_\Gamma^*(C)$, this by $i_C^M : C \to M$ is a proper map, because C is closed in M.
3. We want to say $(i_C^M)^*(x_M) \cong (i_{\tilde{U}}^M)^*(x_M)$ Idea If we consider $Im_t := Im(\alpha_t) \subset \tilde{U}$ and consider the "homotopy"

   $$H_t : Im_t \to M : u \mapsto i_{\tilde{U}}^M(u),$$

   where we would get $H_0 = i_{\tilde{U}}^M$ and $H_1 = i_C^M$.
4. $(i_{\tilde{U}}^M)!(x_M|_{\tilde{U}}) \in \mathscr{H}_\Gamma^*(M)$ is well defined by the fact that the push forward is covariant in open sets, therefore we get this element by the inclusion $i_{\tilde{U}}^M : \tilde{U} \to M$. Aqui requerimos que podamos extender un elemento de cohomologia por cero
5. $(i_C^M)^*(x_M - [(i_{\tilde{U}}^M)!(x_M|_{\tilde{U}})]) = 0 \in \mathscr{H}_\Gamma^*(C)$ this by $i_C^M : C \to M$ is a proper map, because C is closed in M.
6. In the long exact sequence of the pair (M, C) we get that there exist $\tilde{x}_M \in \mathscr{H}_\Gamma^*(M, C) \cong \mathscr{H}_\Gamma^*(M \setminus C)$ such that $\delta(\tilde{x}_M) = x_M - [(i_{\tilde{U}}^M)!(x_M|_{\tilde{U}})] \in \mathscr{H}_\Gamma^*(C)$.

Hence, we define

$$h([M, \partial M, \pi_M, x_M]) = [M \setminus C, \emptyset, \pi_M|_{M \setminus C}, \tilde{x}_M] \in \mathscr{H}_*^{pf}(X \setminus A, \emptyset; \Gamma)$$

□

5 Applications

5.1 Review Baum-Douglas Homology

5.1.1 Analytical K-holomogy

Seccion 4....

Definition 5.1. *$K^a(X)$ is defined as the equivalence clases of 5-tuples $(\mathcal{H}_0, \psi_0, \mathcal{H}_1, \psi_1, T)$, under the relation $\sim$, and where*

- *$\mathcal{H}_i$ is a separable Hilbert space for $i = 0, 1$.*
- *$\psi_i : C(X) \to L(\mathcal{H}_i)$ is a unital algebra $*-$homomorphism.*
- *$T : \mathcal{H}_0 \to \mathcal{H}_1$ is a bounded Fredholm operator with $T \circ \psi_0(f) - \psi_1(f) \circ T$ being a compact operator for any $f \in C(X)$*

$$\begin{tikzcd} \mathcal{H}_0 \arrow[r, "T"] & \mathcal{H}_1 \arrow[d, "\psi_1(f)"] \\ \mathcal{H}_0 \arrow[u, "\psi_0(f)"] \arrow[ur] & \mathcal{H}_1 \arrow[ul] \end{tikzcd}$$

The relation $\sim$ is generated such that

- *Isomorphic 5-tuples are equivalent, where for a pair of cycles $(\mathcal{H}_0, \psi_0, \mathcal{H}_1, \psi_1, T)$ and $(\mathcal{H}'_0, \psi'_0, \mathcal{H}'_1, \psi'_1, T')$, an isomorphism is given by operators $U_i; \mathcal{H}_i \to \mathcal{H}'_i$, for $i = 0, 1$ such that*
  - *$T' = U_1 T U_0^*$.*
  - *$\psi_i(f) = U_i \psi_i(f) U_i^*$.*
- *Direct sum with a trivial object:*

5.1.2 Spin^C-structures

The group $\mathrm{Spin}(n)$ is the double cover of $\mathrm{SO}(n)$, with projection

$$\eta : \mathrm{Spin}(n) \cong \mathbb{Z}_2 \times \mathrm{SO}(n) \to \mathrm{SO}(n).$$

Denote $\epsilon \in Ker(\eta) \cong \mathbb{Z}_2$ the non trivial element and consider

$$\mathrm{Spin}^{\mathrm{c}}(n) = (\mathrm{Spin}(n) \times S^1)/\Gamma,$$

where $\Gamma = \{(\epsilon, -1), (1, 1)\}$ acts on $\mathrm{Spin}(n) \times S^1$ by aclarar accion. For a vector bundle $E \to Y$, consider the fiber bundle $F(E) \to Y$ whose fibre at $p \in Y$ is the set of all vector space bases of E_p (considering Y connected and $\dim_{\mathbb{R}}(E_p) \leq 3$). When $F(E)$ is connected, we say that the vector bundle is orientable, but when $F(E)$ has exactly 2 connected components, we say that the vector bundle $E \to Y$ is orientable, where a choice of one of these two connected components is an orientation for the vector bundle, denoted $F^+(E)$ (where there is an homeomorphism $F^+(E)_p \cong GL^+(n, \mathbb{R})$). Since the cohomology groups of $GL^+(n, \mathbb{R})$ are

$$H^0(GL^+(n, \mathbb{R})) = \mathbb{Z}, \quad H^1(GL^+(n, \mathbb{R})) = 0, \quad H^2(GL^+(n, \mathbb{R})) = \mathbb{Z}_2,$$

then for the inclusion $i_{F^+(E)_p} : F^+(E)_p \to F^+(E)$ we get the cohomology induced map

$$i^*_{F^+(E)_p} : H^2(F^+(E)) \to \mathbb{Z}_2 = \{1, -1\}.$$

Definition 5.2. *A* Spin^c*-structure for a vector bundle* $E \to Y$ *is*

- *An orientation for* E*.*
- *A cohomology class* $u \in (i^*_{F^+(E)_p})^{-1}(-1)$ *for any* $p \in Y$*.*

We say that a smooth manifold M is a Spin^c manifold if there is a Spin^c-structure for the tangent bundle $TM \to M$. If M is a Spin^c manifold with $\partial M \neq \emptyset$, then ∂M is a Spin^c manifold too.

5.1.3 Clutching construction

Let M be a Spin^c manifold, and $H \to M$ be a Spin^c vector bundle with even dimensional fibres. Considering a positive definite symmetric inner product for $H \oplus (M \times \mathbb{R}) \to M$, define

$$\hat{M} := S(H \oplus (M \times \mathbb{R})),$$

which is a even dimensional (A mi me da impar) sphere bundle $\rho : \hat{M} \to M$ and it is a Spin^c manifold (Como se obtiene esta estructura spinc). From [**?**] since $n = \dim_{\mathbb{R}}(H_p)$ is even, $\mathrm{Spin}^c(n)$ has two 1/2-Spin representations (no es $n/2$?) complex vector bundles $H_\pm \to M$. A pairing

$$H \times H_+ \to H_-$$

is given by Clifford multiplication, which defines a vector bundle map

$$\sigma : H_0 := \rho^*(H_+) \to H_1 := \rho^*(H_-),$$

where σ is an isomorphism off the zero section of H(No se que significa eso). Then we can identify

$$\hat{M} = B(H) \bigcup_{S(H)} B(H),$$

where $B(H)$ and $S(H)$ represent unit ball bundle and sphere bundle of H, respectively. With these we construct $(\hat{M}, \hat{H}, \rho)$, where we define

$$\hat{H} = H_0 \cup_\sigma H_1 \to \hat{M}.$$

For each $p \in M$, the restriction of $\hat{H}$ to the even-dimensional sphere $\rho^{-1}(p)$, is the Bott generator vector bundle on $\rho^{-1}(p)$.

5.1.4 Geometrical K-homology

Definition 5.3. *A* K*-cycle on a topological space* X *is a triple* (M, E, ϕ) *such that:*

1. M *is a compact* Spin^C*-manifold without boundary.*
2. $E \to M$ *is a complex vector bundle on* M*.*

3. $\phi : M \to X$ is a continuous map.

Two K-cycles are considered isomorphic if there is a diffeomorphism $h : M \to M'$ such that h preserves the Spin^c-structures, $h^*(E') \cong E$, and $\phi = \phi' \circ h$. The collection of K-cycles is denoted by $\Gamma(X)$.

Definition 5.4. *Define the geometric K-homology of X by $K^{geo}(X) = \Gamma(X)/\sim$, where the relation $\sim$ is an equivalence relation generated by bordism, direct sum and vector bundle modification.*

The vector bundle modification of (M, E, ϕ) by a given smooth Spin^c vector bundle $H \to M$ with even dimensional fibres, is the cycle $(\hat{M}, \hat{H} \otimes \rho^*(E), \phi \circ \rho)$, where

$$\begin{tikzcd} \hat{H} \otimes \rho^*(E) \arrow[d] & E \arrow[d] & \\ \hat{M} \arrow[r, "\rho"] & M \arrow[r, "\phi"] & X \end{tikzcd}$$

$\hat{H}$ is the clutching construction constructed in the previous section.

About $K^{geo}(X)$ we get

1. It is an abelian group with operation given by disjoint union.
2. It define a covariant functor.
3. $K^{geo}(X) = K_0^{geo}(X) \oplus K_1^{geo}(X)$, where $K_0^{geo}(X)$ and $K_1^{geo}(X)$ are the subgroup of cycles (M, E, ϕ) with M even and odd dimensional, respectively.
4. There is a bilinear pairing $K^0(X) \otimes K^{geo}(X) \to K^{geo}(X)$ given by teh cup product $F \cap (M, E, \phi) = (M, E \otimes \phi^* F, \phi)$.

   Example 5.5. *$K_0^{geo}(\{*\}) = \mathbb{Z}$ and $K_1^{geo}(\{*\}) = 0$, by Bott periodicity.*

5.1.5 The isomorphism between Analytical and geometrical K - homologies

For a topological space X, in [**?**] it is defined an isomorphism $\mu : K^g(X) \to K^a(X)$ by

$$\mu(M, E, \phi) = \phi_*(E \cap [D]),$$

where:

1. D is the Dirac operator constructed in the following way: Denote the complexified cotangent bundle by $T^* = \mathbb{C} \otimes_{\mathbb{R}} TM^*$. Denote by $C^\infty(V)$ the vector space of all smooth sections of a smooth vector bundle $V \to M$. Definition of the Dirac operator depends on the parity of $n = dim(M)$:

- If n is even, considering $F_\pm \to M$ the two $\mathrm{Spin}^c(n)$ smooth complex vector bundles given by the Spin^c structure on $TM \to M$Me encantaria poder explicar la manera en que la estructura spin aparece en las estructuras spinc. , the Dirac operator $D : C^\infty(F_+) \to C^\infty(F_-)$ is defined as the composition $D := \gamma_* \circ \nabla$, where
  - $\nabla : C^\infty(F_+) \to C^\infty(T^* \otimes F_+)$ is a connection[5] for $F_+ \to M$.
  - $\gamma : T^* \otimes F_+ \to F_-$ is given by Clifford multiplication, getting $\gamma_*(s) = \gamma \circ s : M \to F_-$ for any $s \in C^\infty(T^* \otimes F_+)$.
- If n is odd, the Spin^c structure on the tangent bundle of M associates to the Spin representation of $\mathrm{Spin}^c(n)$ a smoth complex vector bundle $F \to M$ and in the same fashion of even case, the Dirac operator is defined as the composition:

$$D = \gamma_* \circ \nabla : C^\infty(F) \to C^\infty(T^* \otimes F) \to C^\infty(F).$$

In both cases and for any $E \to M$ smooth vector bundle, choosing a connection for $F \otimes E \to M$, the operator $D \otimes I_E : C^\infty(\widetilde{F} \otimes E) \to C^\infty(\overline{F} \otimes E)$ is the composition

$$D \otimes I_E = \gamma_* \circ \nabla : C^\infty(\widetilde{F} \otimes E) \to C^\infty(T^* \otimes \overline{F} \otimes E) \to C^\infty(\overline{F} \otimes E)$$

where $\widetilde{F} = F_+$ and $\overline{F} = F_-$ in even case, and $\tilde{F} = \overline{F} = F$ in the odd case.

2. Following Section 5 in [BD82], for a Dirac Operator[6] $[D] \in K^a(M)$ is determinded as

$$[D] := (L^2(E_0), \psi_0, L^2(E_1), \psi_1, V),$$

where for $i = 0, 1$:

- For the vector bundles $E_i \to X$, then $L^2(E_i)$ is the completion of $C^\infty(E_i)$ respect to the inner product defined from a choosed measure μ on M, and Hermitian structures on E_i as follows:

$$\langle u, v \rangle = \int_M \langle u(p), v(p) \rangle \, d\mu, \qquad u, v \in C^\infty(E_i).$$

- $V : L^2(E_0) \to L^2(E_1)$ is the partial isometry part in the polar decomposition of the closable operator $D : L^2(E_0) \to L^2(E_1)$ (Buscar referencia para garantizar su existencia y unicidad), being this a bounded operator.

[5] A $\mathbb{C}$-linear map such that $\nabla(f \cdot s) = df \otimes s + f \cdot \nabla(s)$ for $f \in C^\infty(M)$, $s \in C^\infty(F_+)$, where $d : C^\infty(\wedge^j T^*) \to (\wedge^{j+1} T^*)$, is the de Rham operator.

[6] In [BD82], the formulation is in general for any elliptic pseudo differential operator.

- The maps ψ_i are constructed in the following way

$$C(X) \xrightarrow{\psi_i} \mathrm{L}(L^2(E_i))$$

$$f \longmapsto \psi_i(f) : L^2(E_i) \to L^2(E_i) : u \mapsto [\psi_i(f)](u)$$

(Seria chevere ver porque esta es cuadrado integrable) such that for any $e \in E_i$, then

$$[\psi_i(f)](u) : E_i \to E_i : e \mapsto ([\psi_i(f)](u))(e) := f(\rho_{E_i}(e)) \cdot u(e)$$

3. The cap product is defined by ... (Seccion 5, y los profes recomiendan leer el producto de kasparov)

4. $\phi_* : K^a(M) \to K^a(X)$ is the induced map by $\phi : M \to X$.

5.2 Equivariant K-Homology

Definition 5.6 ([BHS07], p. 12)**.** *A complex spinor bundle for V, where $V \to M$ being an euclidean vector bundle over M of rank p, is a p-multigraded Dirac bundle S_V which is locally isomorphic to the trivial bundle with fiber the complex Clifford algebra $\mathbb{C}_p$, where:*

1. *A p-multigraded Dirac bundle structure on V* (see [BHS07], p. 6) *is a smooth $\mathbb{Z}_2$-graded, Hermitian vector bundle S over M together with the following data:*

   (a) *An $\mathbb{R}$-linear morphism of vector bundles $V \to End(S)$ which associates to each vector $v \in V_x$ a skew-adjoint, odd-graded endomorphism $u \mapsto v \cdot u := v(u)$ of S_x, in such a way that*

   $$v \cdot v \cdot u = -||v||^2 u.$$

   (b) *A family of skew-adjoint, odd-graded endomorphism $\varepsilon_1, ..., \varepsilon_p$ of S such that*

   $$\varepsilon_j = -\varepsilon_j^*, \quad \varepsilon_j^2 = -1, \quad \varepsilon_i\varepsilon_j + \varepsilon_j\varepsilon_i = 0 (i \neq j),$$

   and such that each ε_j commutes with each operator $u \mapsto v \cdot u$.

If M is a smooth manifold (possibly with boundary) then by a spin^c structure on M we mean a pair consisting of a Riemannian metric on M and a complex spinor bundle S_M for TM. Then for each $x \in M$, there exist an open $x \in U_x \subseteq M$ with the following diagram

$$\begin{tikzcd}
S_M|_{U_x} \cong U_x \times \mathbb{C}_p \arrow[d] & S_M \arrow[dr] & TM \arrow[r] \arrow[d] & End(S_M) \arrow[dl] \\
U_x \arrow[rr, hook] & & M &
\end{tikzcd}$$

A Γ-spin^c manifold is a manifold with a spin^c-structure given in terms of a complex spinor bundle S_M for TM with a Γ-action listed to and compatible with all the structure.Ser más específico

Definition 5.7 ([BOOSW10], p. 5)**.** *The equivariant K-homology respect to a discrete group Γ[7] of a Γ-CW pair (X, Y), denoted $K_*^{\Gamma,geom}(X, Y)$ is the set of isomorphism classes of* **cycles** *identified by a equivalence* **relation** *$\sim$, those defined as follows:*

1. *A* **cycle** *is a triple (M, E, f), where*
   - *M is a compact smooth Γ-spin^c manifold.*
   - *E is a Γ-equivariant Hermitian vector bundle on M.*
   - *$f : M \to X$ is a continuous Γ-equivariant map such that $f(\partial M) \subset Y$.*
2. *An isomorphism between two cycles (M, E, f) and (M', E', f') is a smooth Γ-map $\phi : M \to M'$ such that*
   - *(a) push or pull back spin^c structure.*
   - *(b) $\phi^*(E') = E$.*
   - *(c) $f = f' \circ \phi$.*
3. *The equivalence* **relation** *generated by the following conditions:*
   - $(M, E, f) \sqcup (M, E', f) \sim (M, E \oplus E', f)$.

$$\begin{tikzcd}
E \arrow[r, hook] \arrow[dr] & E \oplus E' \arrow[d] & E' \arrow[l, hook'] \arrow[dl] & \\
& M \arrow[r, "f"] & X &
\end{tikzcd}$$

   - *$(M, E, f) \sim (M', E', f')$ if there is a bordism between (M, E, f) and $-(M', E', f')$[8], where a bordism beetwen K-cycles over (X, Y), identified as $M_+ = g^{-1}[1, \infty)$ and $M_+ = g^{-1}(-\infty, -1]$, is the following bunch of data:*
     - *A smooth, compact Γ-manifold L, equipped with a Γ-spin^c-structure.*
     - *A smooth, Hermitian Γ-vector bundle F over L.*
     - *A continuous Γ-map $\Phi : L \to X$.*

[7] This is defined in general for a compact Lie group G, see [BOOSW10].

[8] By $-(M', E', f')$ we mean the K-cycle $(-M', E', f')$ where $-M'$ is the manifold M' equipped with the opposite spin^c structure of M'.

– *A smooth Γ-invariant map $g : \partial L \to \mathbb{R}$ for which ± 1 are regular values, and for wich $\Phi[g^{-1}[-1,1]] \subseteq Y$.*

$$\begin{tikzcd}
\mathbb{R} & F \arrow[d] & \\
\partial L \arrow[u, "g"] \arrow[r, hook] & L \arrow[r, "\Phi"] & X \\
g^{-1}([-1,1]) \arrow[u, hook] \arrow[ur, hook, "i"] \arrow[rr, "\Phi\circ i"'] & & Y \arrow[u, hook]
\end{tikzcd}$$

then we mean that between (M, E, f) and $-(M', E', f')$ there exist a bordism such that $g^{-1}(\{1\}) = M$, $g^{-1}(\{-1\}) = -M'$, $i^*F \cong E \sqcup E' \to g^{-1}(\{-1,1\}) = M \sqcup (-M')$ Revisar

- *$(M, E, f) \sim (Z, F \otimes \pi^* E, f \circ \pi)$, as in the following diagram*

$$\begin{tikzcd}
F \otimes \pi^* E \arrow[d] & E \arrow[d] & \\
Z \arrow[r, "\pi"] & M \arrow[r, "f"] & X
\end{tikzcd}$$

being the later the modification of a Γ-spin^c bundle $W \to M$, where:

(a) Z is the unit sphere bundle of the bundle $\pi : Z \subset 1 \oplus W \to M$.

$$\begin{tikzcd}
Z \arrow[r, hook] \arrow[dr, "\pi"'] & 1 \oplus W \arrow[d] \\
& M
\end{tikzcd}$$

(b) F is obtained as $\pi^* S^*_{W,+}$ over the northern hemisphere[9] D^+ of Z and $\pi^* S^*_{W,+}$ over the southern hemisphere D^- of Z by gluing along the intersection, the unit sphere bundle of W, using Clifford multiplication with the respective vector of W.

$$\begin{tikzcd}
\pi^* S^*_{W,\pm} \arrow[d] & S^*_{W,\pm} \arrow[d] & \Rightarrow & F \cong \pi^* S^*_{W,+} \sqcup_{t=0} \pi^* S^*_{W,-} \arrow[d] \\
D^\pm \arrow[r, "\pi"'] & M & & Z \cong D^+ \sqcup_{t=0} D^-
\end{tikzcd}$$

5.3 Periodic Cyclic Homology

5.4 Pushforward homology as a quotient group

Consider the free abelian group generated by the collection of isomorphic cycles $(M, \partial M, \pi_M, x_M)$ related to (X, A) just like in Definition **??**, denoted by $\mathbb{Z}_{\mathscr{H}^*_\Gamma}[Cyc]$, and consider the subgroup $\mathcal{R}_{\mathscr{H}^*_\Gamma}[Cyc]$ generated by:

[9] The notion of Hemisphere has sense identifying Z as the pairs $(t, w) \in 1 \oplus W$ where $t \in [-1, 1]$ and $t^2 + |w|^2 = 1$, and the hemispheres are defined by the sign of t.

- $[M \sqcup N, \partial M \sqcup \partial N, \pi_M \sqcup \pi_N, x_M \oplus x_N] - [M, \partial M, \pi_M, x_M] - [N, \partial N, \pi_N, x_N]$.
- $[M, \partial M, \pi_N \circ f, x_M] - [N, \partial N, \pi_N, f!(x_M)]$, where $f : (M, \partial M) \to (N, \partial N)$ is a proper Γ-equivariant, orientation preserving smooth map.
- $[M, \partial M, \pi_M, x_M + x'_M] - [M, \partial M, \pi_M, x_M] - [M, \partial M, \pi_M, x'_M]$.

6 Assembly map

In [RWW23] there exist a way to relate cycles using a well defined group in the case of the pair $(\{*\}, \{*\})$ (wich is not a Γ proper pair).

7 Generalized $\mathscr{H}^*_\Gamma$-oriented bundles

For a fixed $\mathscr{H}^*_\Gamma$-cohomology, we will consider the $\mathscr{H}^*_\Gamma$-orientation of Γ-vector bundles as a class of Γ-vector bundles such that the class its closed by

1. smooth orientation inverse (classical orientation in the de Rham cohomology);
2. Witney sum;
3. Pull backs;

such that for each of the vector bundle $E \to M$ of this class there exist an isomorphism $\tau_{E,M} : \mathscr{H}^*_\Gamma(M) \to \mathscr{H}^*_\Gamma(E)$ such that

1. For $E = M \times 0$, $\tau_{E,M} = Id_{\mathscr{H}^*(M)}$.
2. For $f : N \to M$, and an $\mathscr{H}$-vector bundle $E \to N$, $f! \circ \tau_{f^*E,M} = \tau_{E,N} \circ f!$.
3. For a pair of vector bundles $E \to M$ and $F \to M$, and a vector bundle V over E and F at the same time, such that $\rho_E \circ \rho_{V,E} = \rho_F \circ \rho_{V,F}$, then $\tau_{V,E} \circ \tau_{E,M} = \tau_{V,F} \circ \tau_{F,M}$.

Proposition 7.1. *If $E = -F$ (opposite in the sense of de Rham cohomology), then $\tau_{E,M} = -\tau_{F,M}$.*

8 Examples

- $\Gamma = \{e\}$.
- $\mathscr{H}^{pf}_*(M, \partial M; \Gamma) \cong \mathscr{H}^*_\Gamma(M, \partial M)$.

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